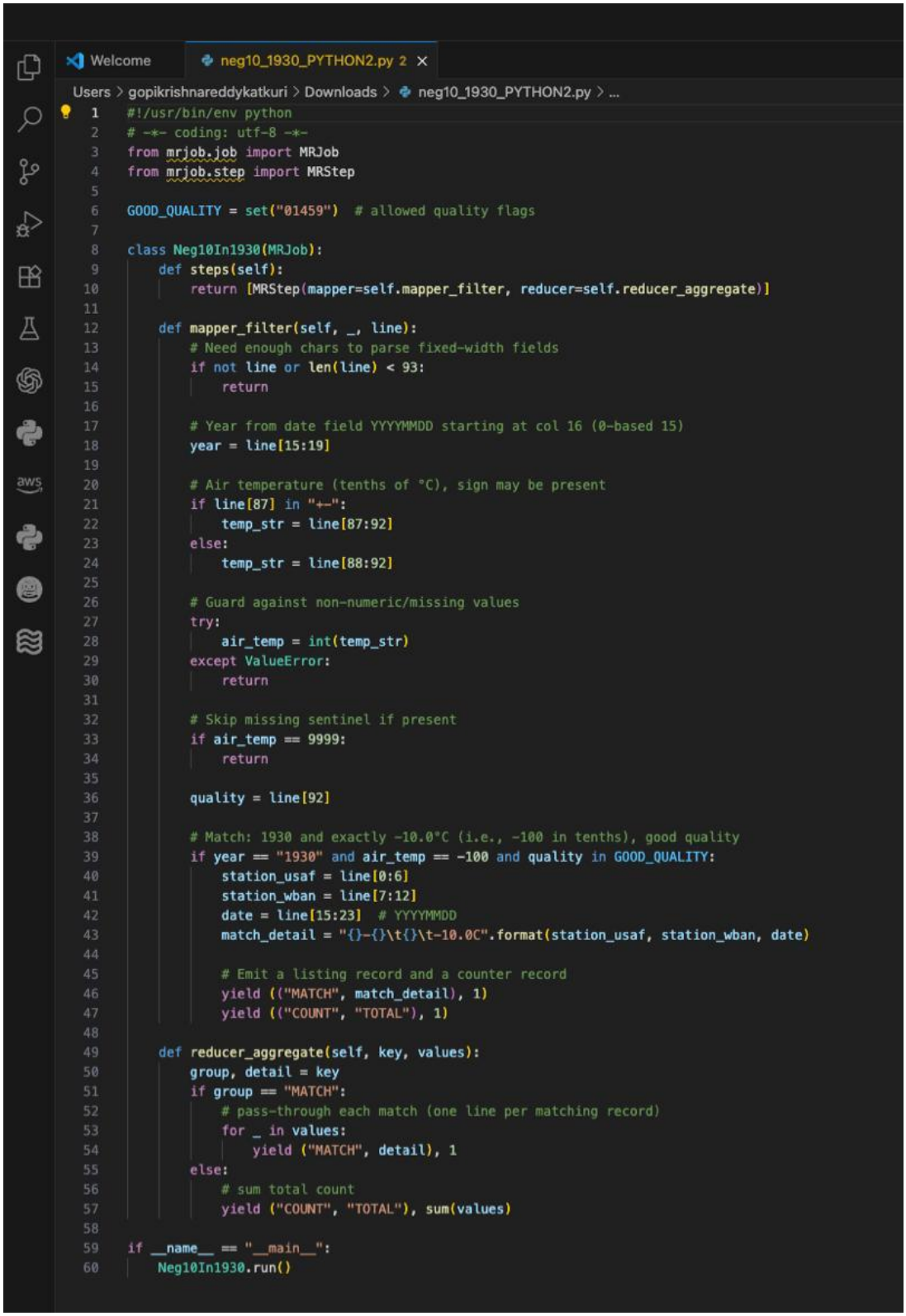
Homework Assignment 2

Q1:

Write a Python MRJob program to retrieve all temperatures of -10°C in 1930 and count them. Test this program in CSUEB Hadoop by using the input data set of 1930, and type out all the commands in each step of the process and print out a screenshot(s) of all the steps and the final results. Find the similar Java example on Page 273 and 274

Step 1: Make python MRJob file

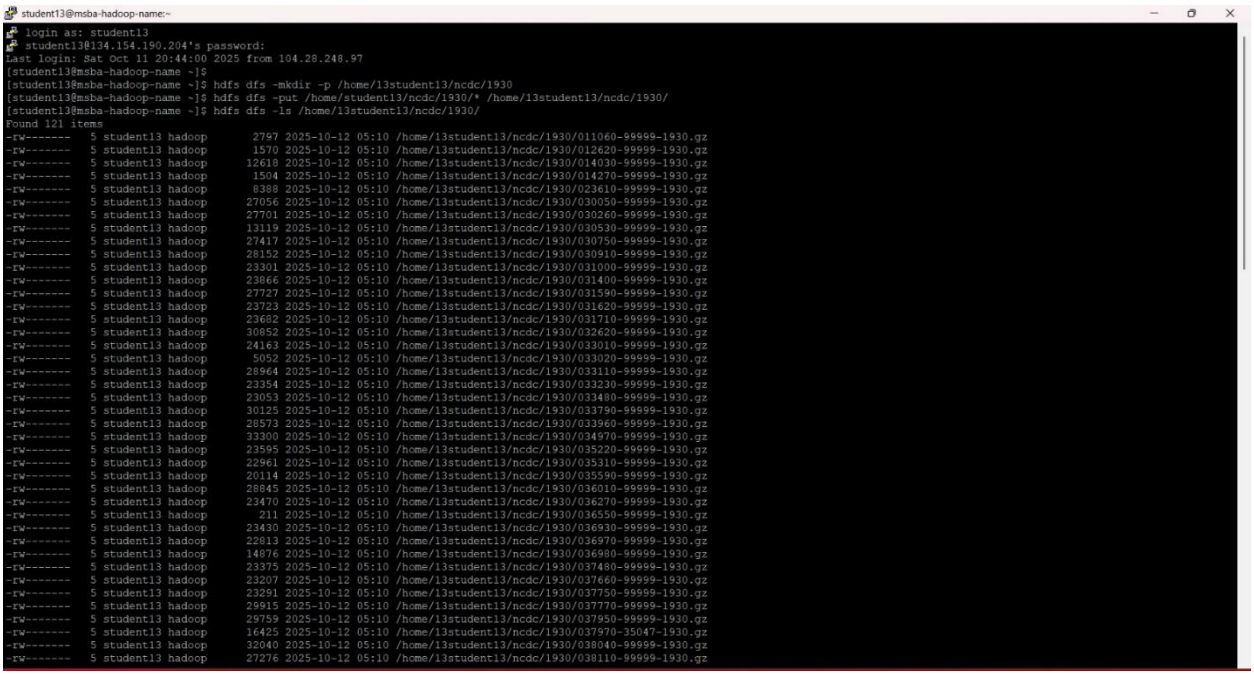
neg10_1930_PYTHON.py



Step 2: Make directory and put the 1930 folder (and files) in HDFS

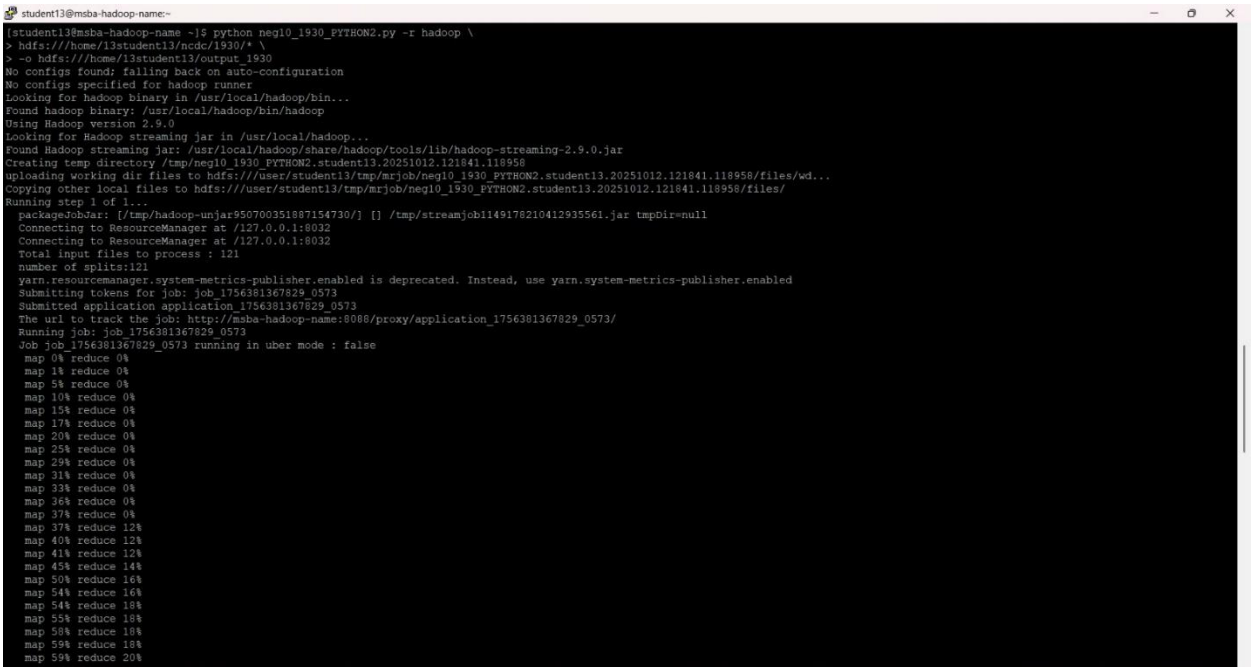
hdfs dfs -mkdir -p /home/13student13/ncdc/1930

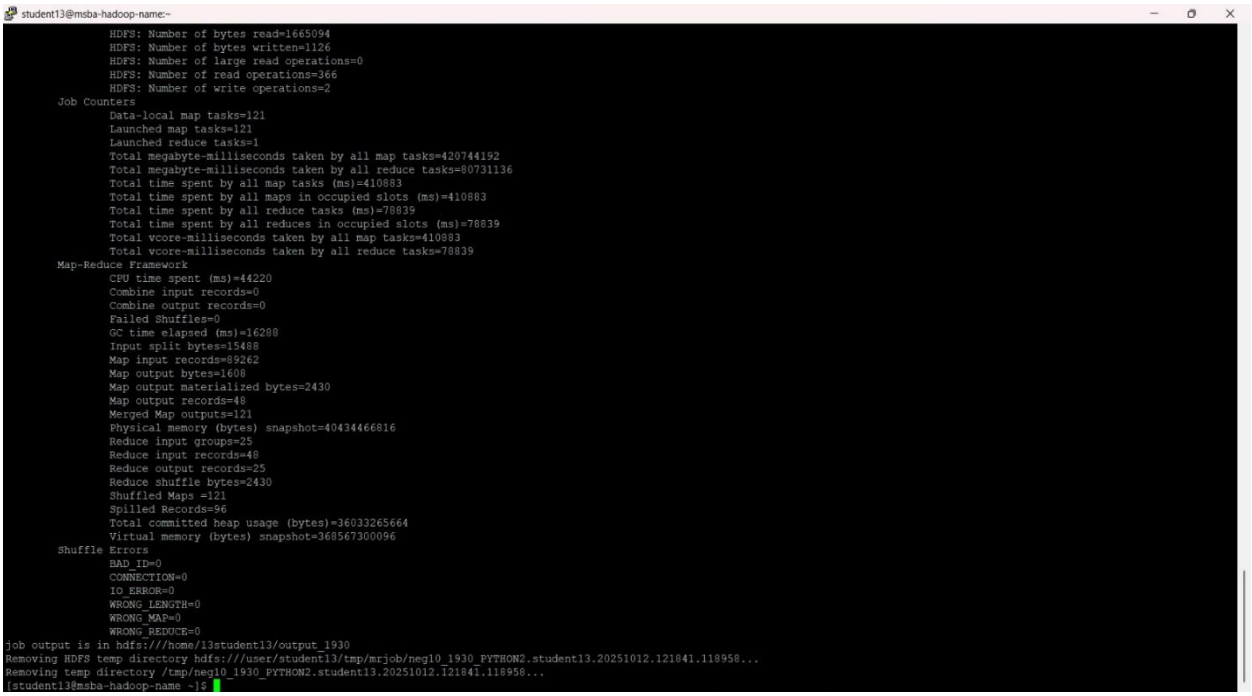
hdfs dfs -put /home/student13/1930/* /home/13student13/ncdc/1930/ hdfs dfs -ls /home/13student13/ncdc/1930



Step 3:Run the neg10_1930_PYTHON2.py file

python neg10_1930_PYTHON2.py -r hadoop \ hdfs:///home/13student13/ncdc/1930/* \ -o hdfs:///home/13student13/output_1930/

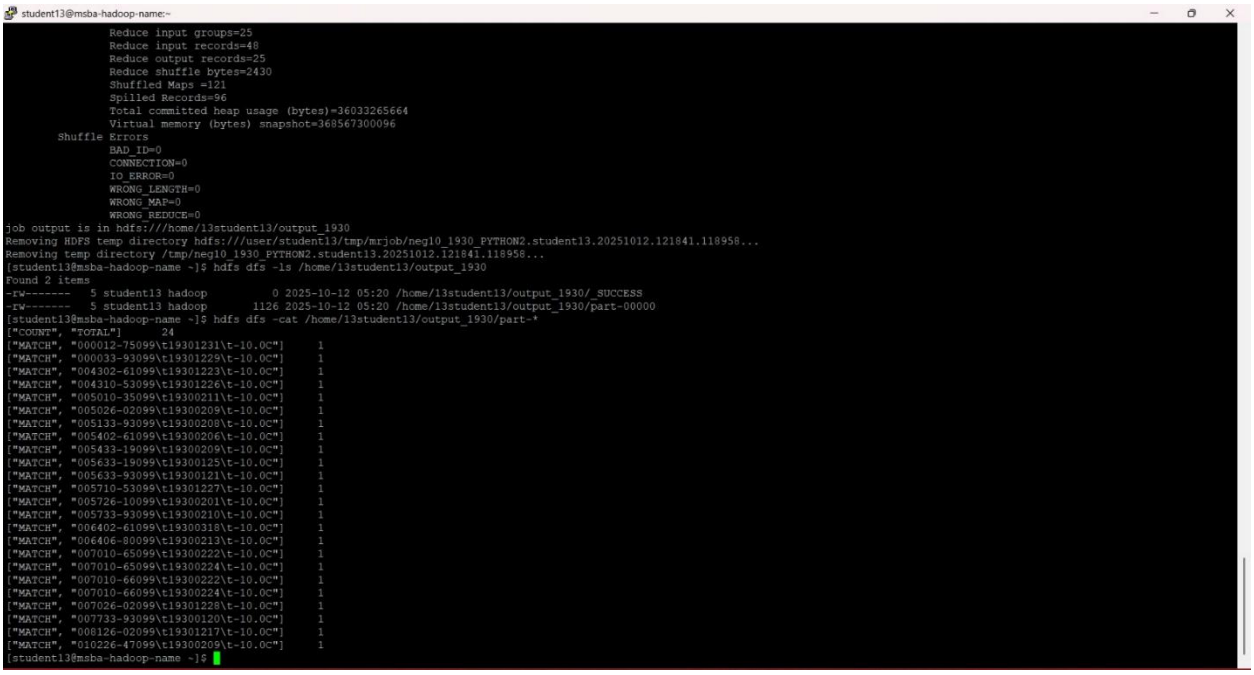




Step 4: Display Output file and the result

hdfs dfs -ls /home/13student13/output_1930/

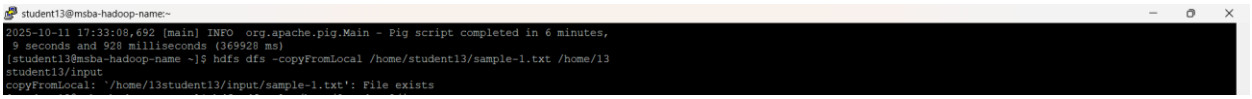
hdfs dfs -cat /home/13student13/output_1930/part-00000



Q2: Work the five examples on pages 384 and 385, that is, from the title “Schemas” to the subtitle “Validation and nulls.” Print out a screenshot(s) for all the steps and the results of these five examples in CSUEB Hadoop.

Copy from server to HDFS and display in HDFS
This file is present in HDFS as *sample-1.txt*)

hdfs dfs -copyFromLocal /home/student13/sample-1.txt /home/13student13/input



hdfs dfs -ls /home/13student13/input



#the bellow screenshot contain all the result output for below commands

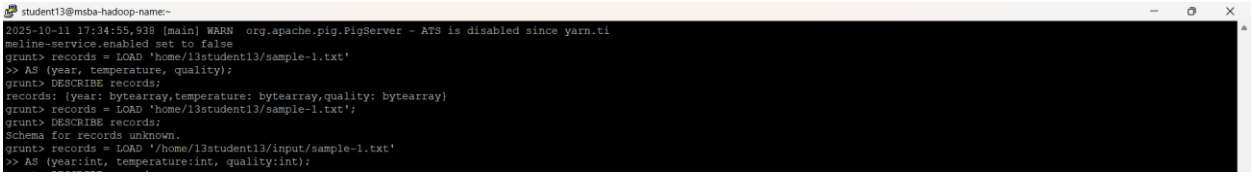
When data type is not defined:

grunt> records = LOAD 'home/13student13/sample-1.txt'
>> AS (year, temperature, quality);

grunt> DESCRIBE records;

When schema is not defined:

grunt> records = LOAD 'home/13student13/sample-1.txt';
grunt> DESCRIBE records;

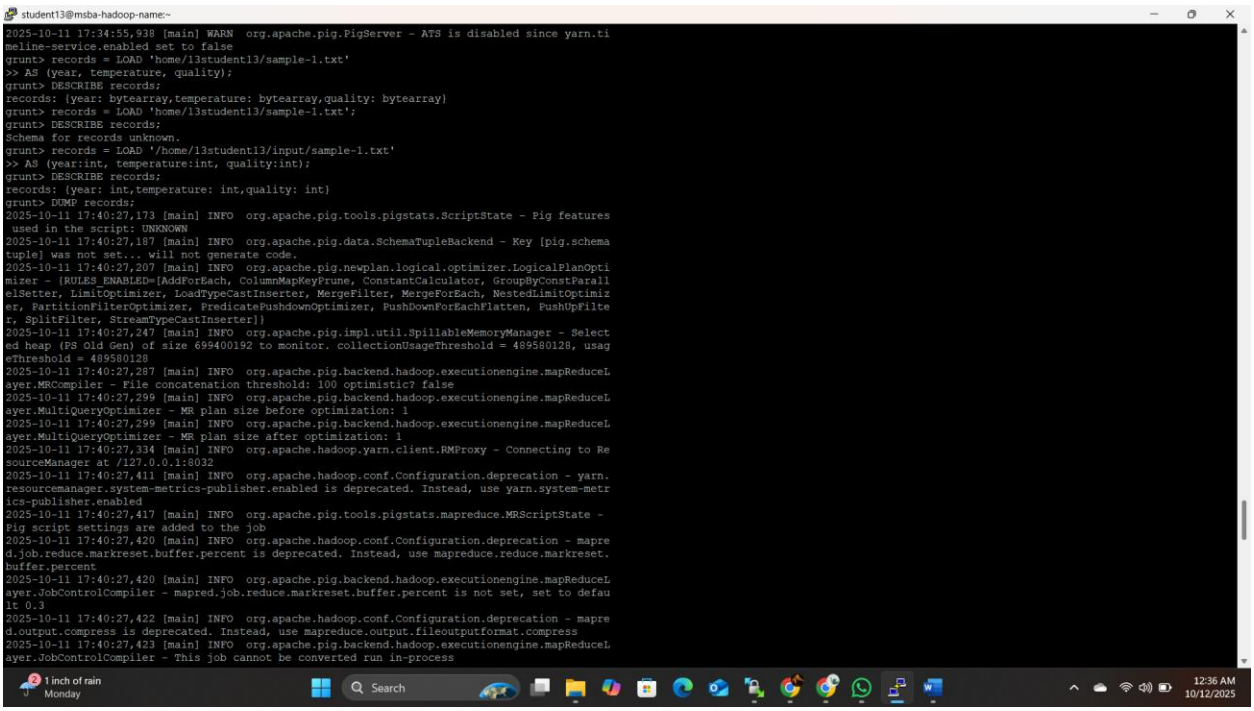


a. Displaying context of sample-1.txt -

Load data from *sample-1.txt* and store in a relation called 'records' and define schema and data type for the records in the file:

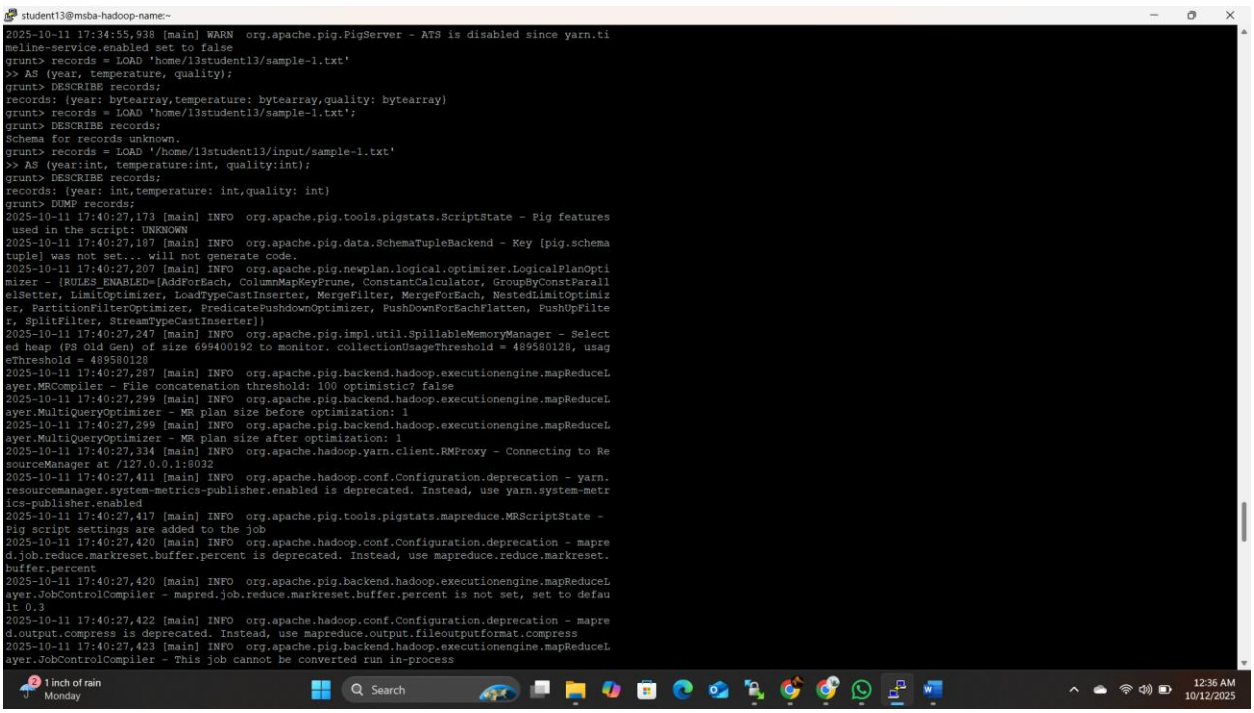
grunt> records = LOAD '/home/13student13/input/sample-1.txt'

>> AS (year:int, temperature:int, quality:int);



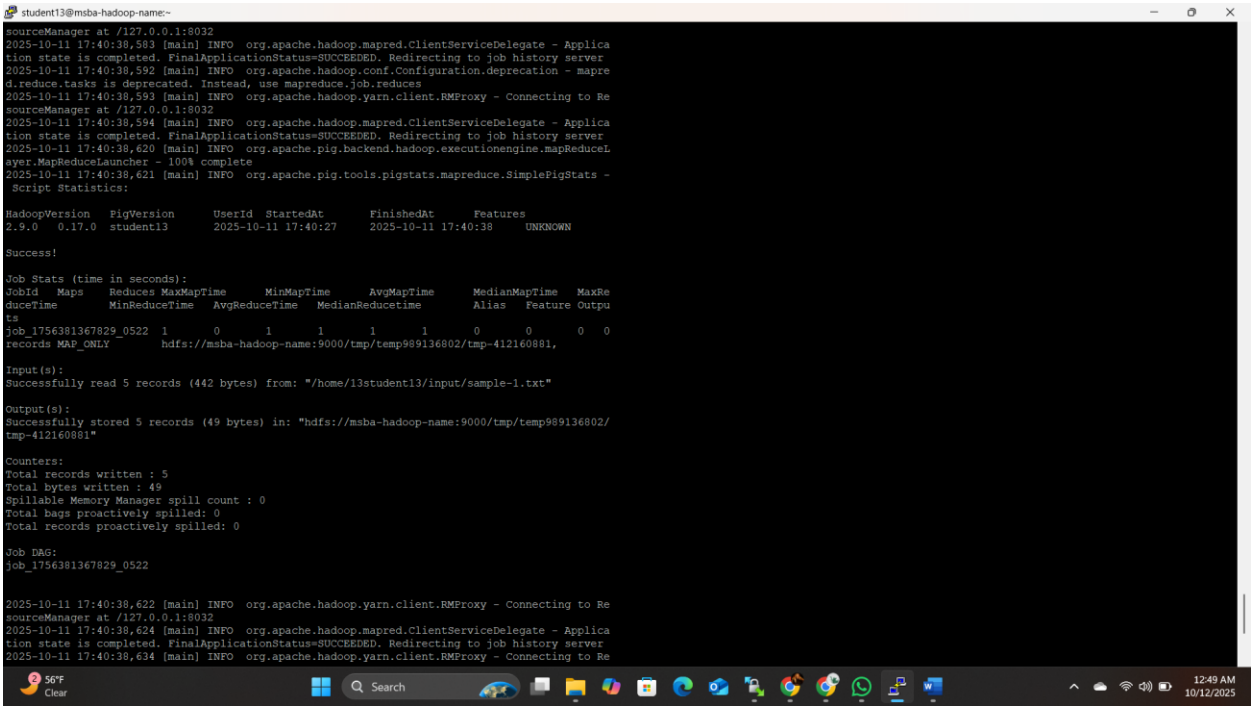
To show the schema:

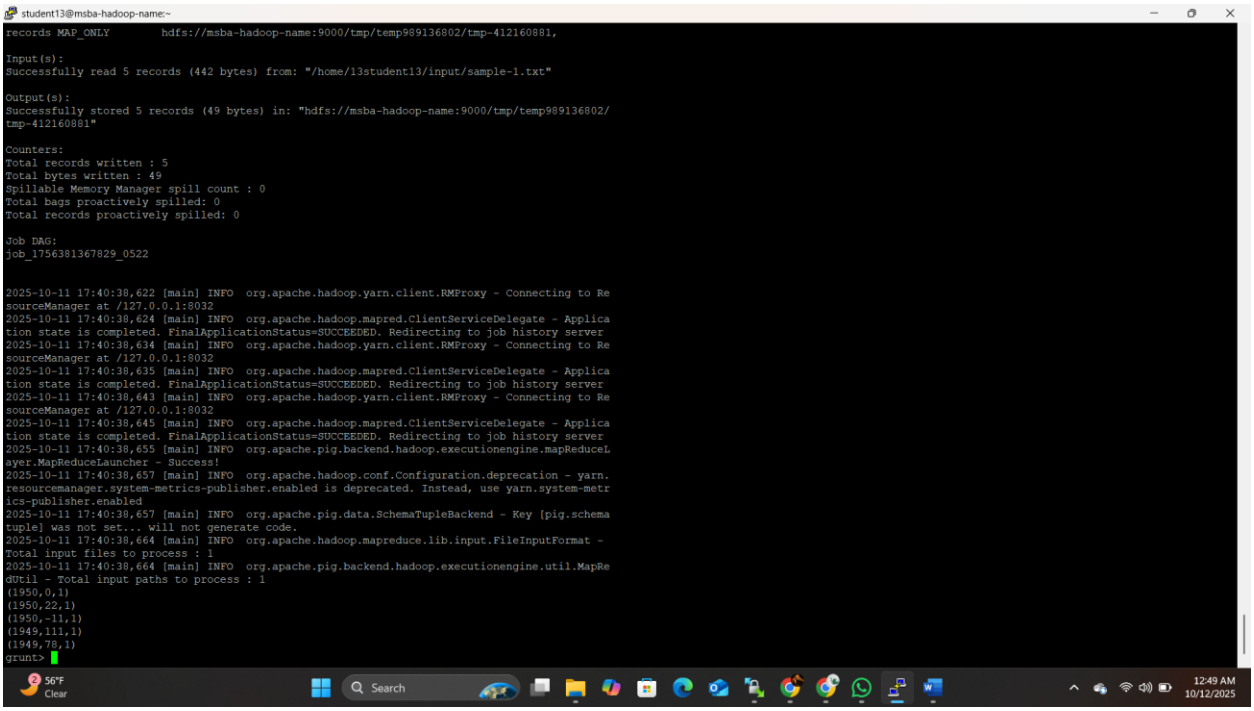
grunt> DESCRIBE records;



Display contents:

grunt> DUMP records



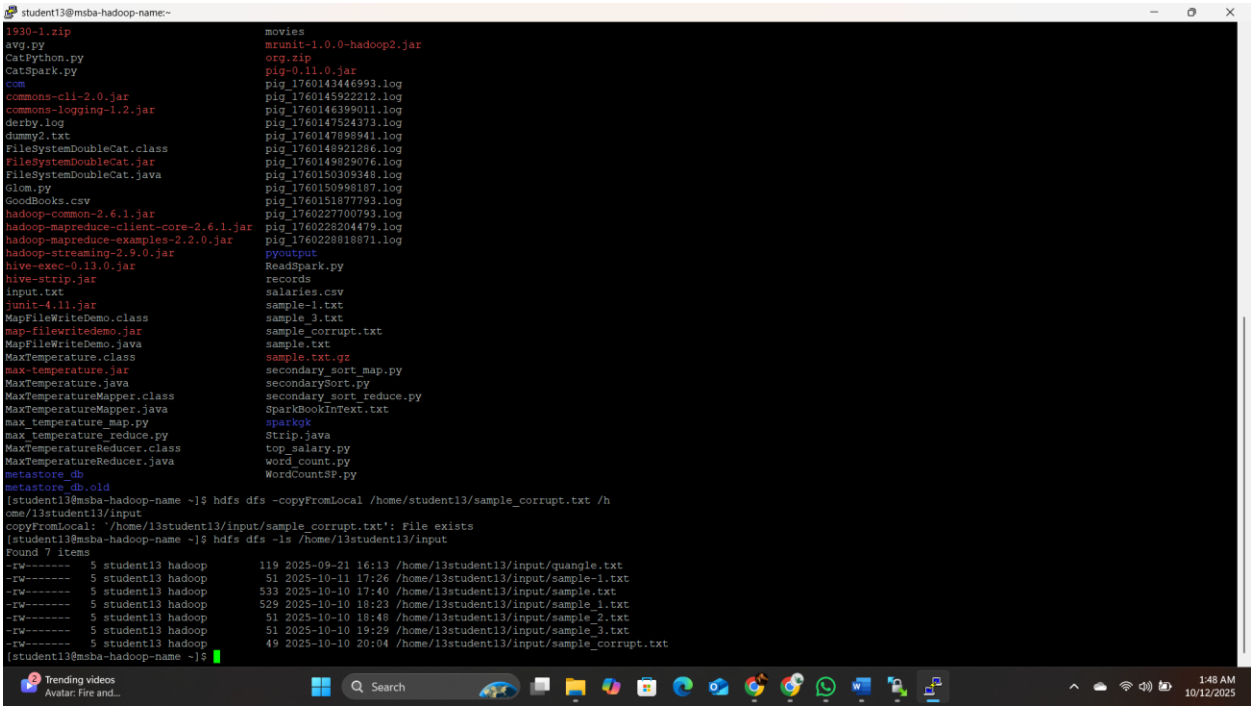


b. Display contents of sample_corrupt.txt -

Copy *sample_corrupt.txt* from server to HDFS and display in HDFS

hdfs dfs -copyFromLocal /home/student13/sample_corrupt.txt /home/13student13/input

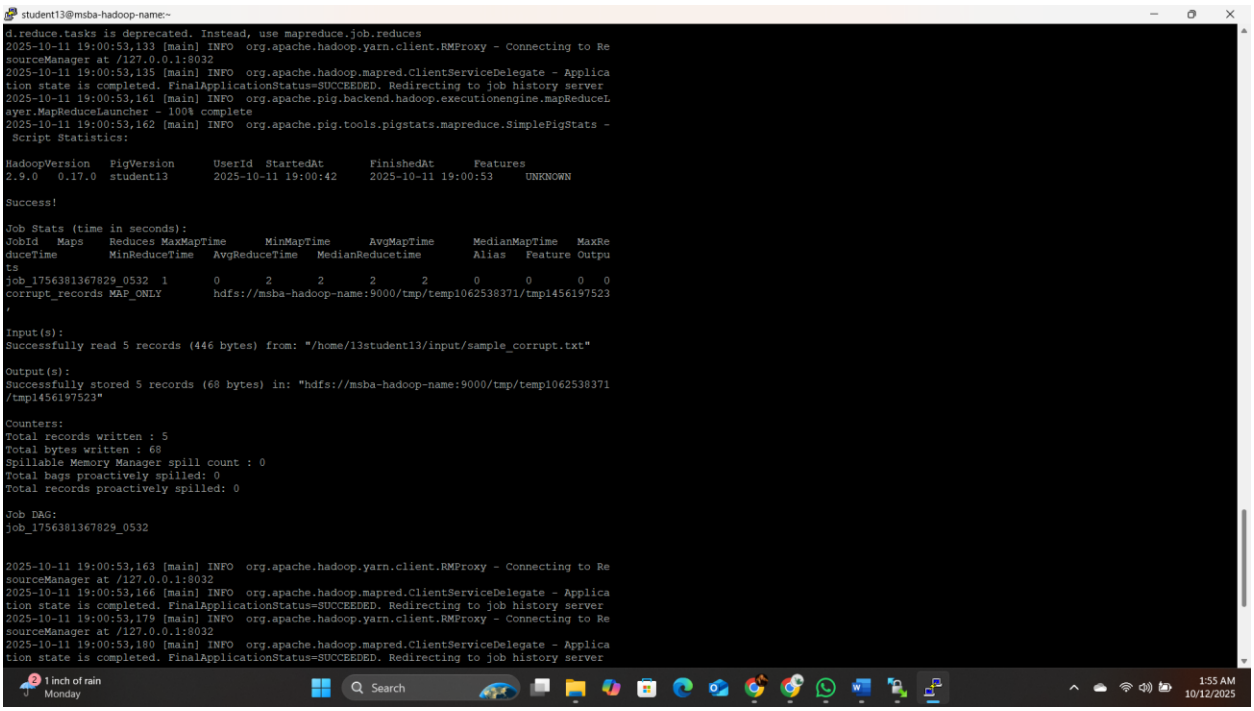
hdfs dfs -ls /home/13student13/input

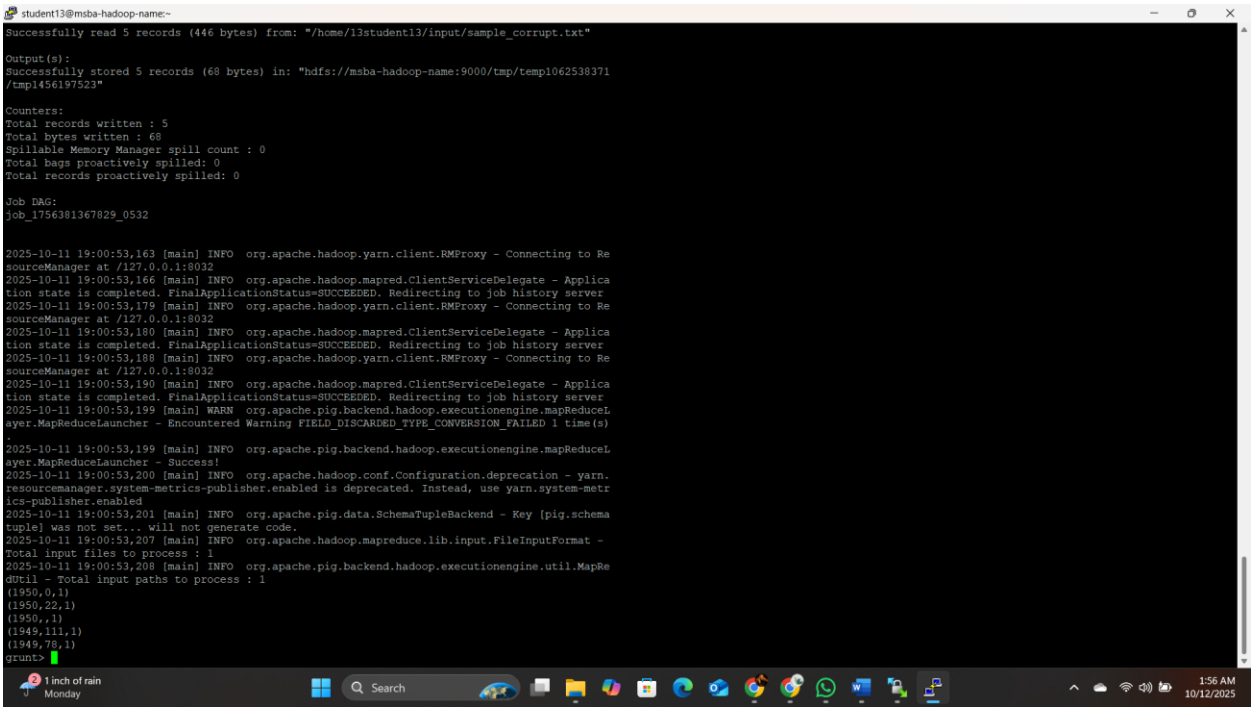


Loading *sample_corrupt.txt*:

grunt> corrupt_records = LOAD '/home/13student13/input/sample_corrupt.txt'
>> AS (year:chararray, temperature:int, quality:int);

grunt> DUMP corrupt_records;

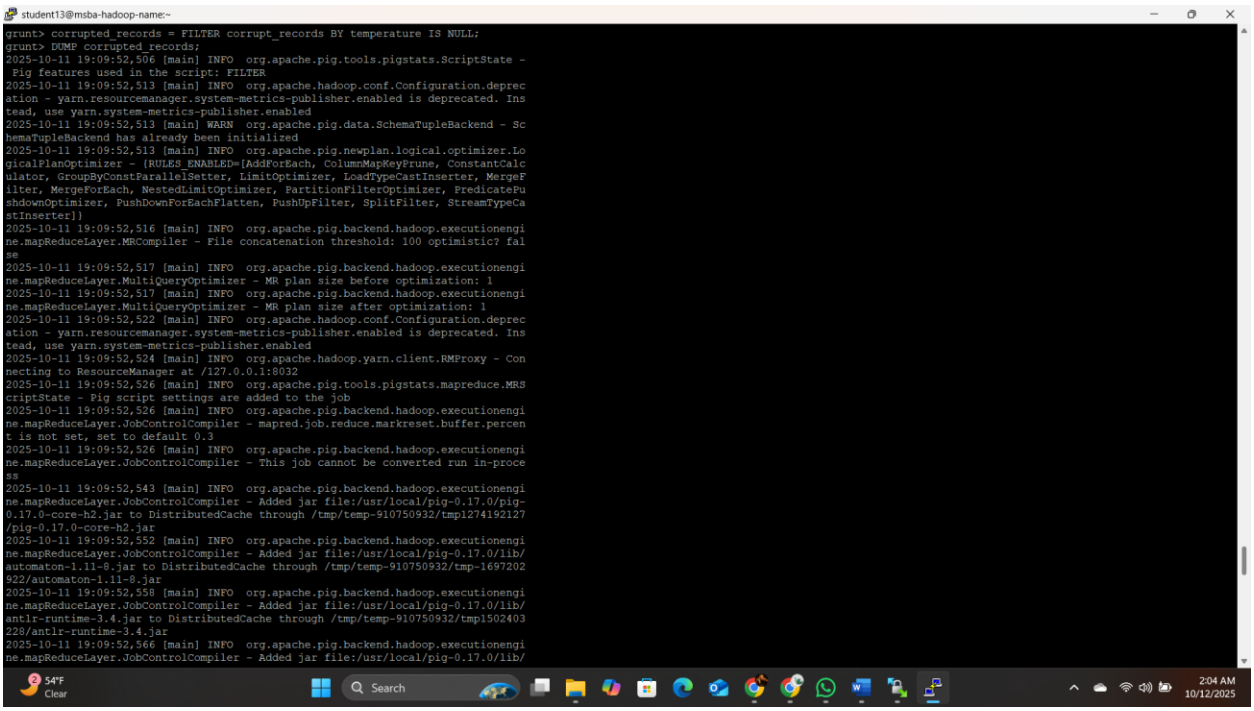

student13@msba-hadoop-name:~
d.reduce.tasks is deprecated. Instead, use mapreduce.job.reduces
2025-10-11 19:00:53,133 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:00:53,135 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:00:53,161 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.MapReduceLauncher - 100% complete
2025-10-11 19:00:53,162 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats -
Script Statistics:
HadoopVersion PigVersion UserId StartedAt FinishedAt Features
2.9.0 0.17.0 student13 2025-10-11 19:00:42 2025-10-11 19:00:53 UNKNOWN
Success!
Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxRe
duceTime MinReduceTime AvgReduceTime MedianReducetime Alias Feature Outpu
ts
job_1756381367829_0532 1 0 2 2 2 2 0 0 0 0
corrupt_records MAP_ONLY hdfs://msba-hadoop-name:9000/tmp/temp1062538371/tmp1456197523
,
Input(s):
Successfully read 5 records (446 bytes) from: "/home/13student13/input/sample_corrupt.txt"
Output(s):
Successfully stored 5 records (68 bytes) in: "hdfs://msba-hadoop-name:9000/tmp/temp1062538371
/tmp1456197523"
Counters:
Total records written : 5
Total bytes written : 68
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0
Job DAG:
job_1756381367829_0532
2025-10-11 19:00:53,163 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:00:53,166 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:00:53,179 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:00:53,180 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
1 inch of rain
Monday
Search
1:55 AM
10/12/2025

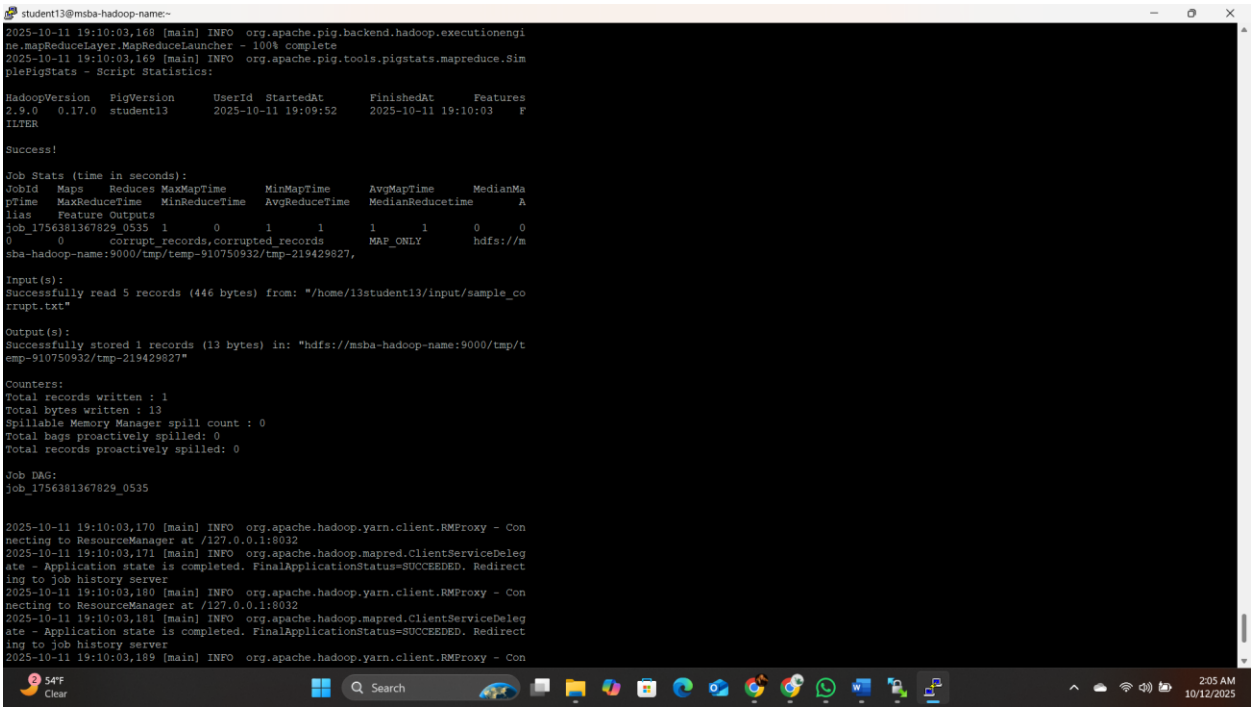

student13@msba-hadoop-name:~
Successfully read 5 records (446 bytes) from: "/home/13student13/input/sample_corrupt.txt"
Output(s):
Successfully stored 5 records (68 bytes) in: "hdfs://msba-hadoop-name:9000/tmp/temp1062538371
/tmp1456197523"
Counters:
Total records written : 5
Total bytes written : 68
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0
Job DAG:
job_1756381367829_0532
2025-10-11 19:00:53,163 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:00:53,166 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:00:53,179 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:00:53,180 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:00:53,188 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:00:53,190 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:00:53,199 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 1 time(s)
.
2025-10-11 19:00:53,199 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.MapReduceLauncher - Success!
2025-10-11 19:00:53,200 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.
resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metr
ics-publisher.enabled
2025-10-11 19:00:53,201 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schema
tuple] was not set... will not generate code.
2025-10-11 19:00:53,207 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat -
Total input files to process : 1
2025-10-11 19:00:53,208 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRe
dUtil - Total input paths to process : 1
(1950,0,1)
(1950,22,1)
(1950,,1)
(1949,111,1)
(1949,78,1)
grunt>
1 inch of rain
Monday
Search
1:56 AM
10/12/2025

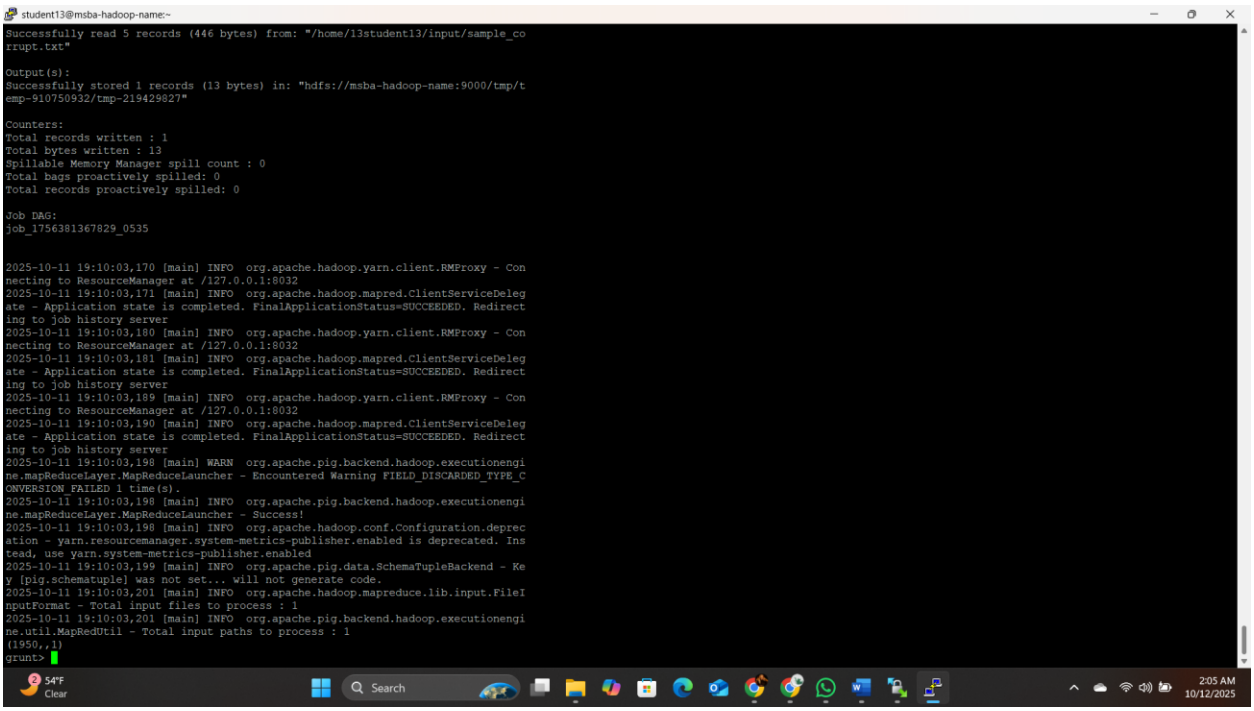
c. Filtering corrupt records where temperature information is not present -

```
grunt> corrupted_records = FILTER corrupt_records BY temperature IS NULL;
```

```
grunt> DUMP corrupted_record
```


student13@msba-hadoop-name:~
grunt> corrupted_records = FILTER corrupt_records BY temperature IS NULL;
grunt> DUMP corrupted_records;
2025-10-11 19:09:52,506 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: FILTER
2025-10-11 19:09:52,513 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:09:52,513 [main] WARN org.apache.pig.data.SchemaTupleBackend - SchemaTupleBackend has already been initialized
2025-10-11 19:09:52,513 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2025-10-11 19:09:52,516 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2025-10-11 19:09:52,517 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2025-10-11 19:09:52,517 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2025-10-11 19:09:52,522 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:09:52,524 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:09:52,526 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2025-10-11 19:09:52,526 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2025-10-11 19:09:52,526 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2025-10-11 19:09:52,543 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/pig-0.17.0-core-h2.jar to DistributedCache through /tmp/temp-910750932/tmp1274192127/pig-0.17.0-core-h2.jar
2025-10-11 19:09:52,552 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp-910750932/tmp-1697202922/automaton-1.11-8.jar
2025-10-11 19:09:52,558 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp-910750932/tmp1502403228/antlr-runtime-3.4.jar
2025-10-11 19:09:52,566 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/
54°F Clear
Search
2:04 AM 10/12/2025


student13@msba-hadoop-name:~
2025-10-11 19:10:03,168 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2025-10-11 19:10:03,169 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:
HadoopVersion PigVersion UserId StartedAt FinishedAt Features
2.9.0 0.17.0 student13 2025-10-11 19:09:52 2025-10-11 19:10:03 FILTER
Success!
Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReducetime Alias Feature Outputs
job_1756381367829_0535 1 0 1 1 1 1 0 0 0 0 corrupt_records,corrupted_records MAP_ONLY hdfs://msba-hadoop-name:9000/tmp/temp-910750932/tmp-219429827,
Input(s):
Successfully read 5 records (446 bytes) from: "/home/13student13/input/sample_corrupt.txt"
Output(s):
Successfully stored 1 records (13 bytes) in: "hdfs://msba-hadoop-name:9000/tmp/temp-910750932/tmp-219429827"
Counters:
Total records written : 1
Total bytes written : 13
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0
Job DAG:
job_1756381367829_0535
2025-10-11 19:10:03,170 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:10:03,171 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:10:03,180 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:10:03,181 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:10:03,189 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Con
54°F Clear
Search
2:05 AM 10/12/2025



d. Grouping the number of corrupt records -

grunt> group_corrupt = GROUP corrupted_records ALL;

grunt> all_grouped = FOREACH group_corrpt GENERATE group, COUNT(corrupted_records);

grunt> DUMP all_grouped;

```
Details at logfile: /home/student13/pig_1760234699361.log
grunt> group_corrupt = GROUP corrupted_records ALL;
grunt> all_grouped = FOREACH group_corrupt GENERATE group, COUNT(corrupted_records);
grunt> DUMP all_grouped;
2025-10-11 19:18:52,806 [main] INFO  org.apache.pig.tools.pigstats.ScriptState - Pig features
 used in the script: GROUP_BY,FILTER
2025-10-11 19:18:52,812 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - yarn.
resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metr
ics-publisher.enabled
2025-10-11 19:18:52,813 [main] INFO  org.apache.pig.data.SchemaTupleBackend - Key [pig.schema
tuple] was not set... will not generate code.
2025-10-11 19:18:52,813 [main] INFO  org.apache.pig.newplan.logical.optimizer.LogicalPlanOpti
mizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParall
elSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimiz
er, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilte
r, SplitFilter, StreamTypeCastInserter]}
2025-10-11 19:18:52,819 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2025-10-11 19:18:52,821 [main] INFO  org.apache.pig.backend.hadoop.executionengine.util.Combi
nerOptimizerUtil - Choosing to move algebraic foreach to combiner
2025-10-11 19:18:52,829 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.MultiQueryOptimizer - MR plan size before optimization: 1
2025-10-11 19:18:52,829 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.MultiQueryOptimizer - MR plan size after optimization: 1
2025-10-11 19:18:52,836 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation - yarn.
resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metr
ics-publisher.enabled
2025-10-11 19:18:52,837 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:18:52,839 [main] INFO  org.apache.pig.tools.pigstats.mapreduce.MRScriptState -
Pig script settings are added to the job
2025-10-11 19:18:52,839 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to defau
lt 0.3
2025-10-11 19:18:52,839 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2025-10-11 19:18:52,840 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.JobControlCompiler - Setting Parallelism to 1
2025-10-11 19:18:52,840 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.JobControlCompiler - This job cannot be converted run in-process
2025-10-11 19:18:52,868 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/pig-0.17.0-core-h2.jar to Dist
ributedCache through /tmp/temp-910750932/tmp-259182694/pig-0.17.0-core-h2.jar
2025-10-11 19:18:52,874 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/automaton-1.11-8.jar to Di
stributedCache through /tmp/temp-910750932/tmp680155601/automaton-1.11-8.jar
2025-10-11 19:18:53,281 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/antlr-runtime-3.4.jar to D
```

DEN - NYJ
In 5 hours
Search
2:18 AM
10/12/2025

student13@msba-hadoop-name:~

```
2025-10-11 19:19:08,393 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:19:08,395 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:19:08,405 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:19:08,406 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:19:08,413 [main] INFO  org.apache.pig.backend.hadoop.executionengine.mapReduceL
ayer.MapReduceLauncher - 100% complete
2025-10-11 19:19:08,414 [main] INFO  org.apache.pig.tools.pigstats.mapreduce.SimplePigStats -
 Script Statistics:

HadoopVersion   PigVersion      UserId  StartedAt       FinishedAt      Features
2.9.0   0.17.0  student13       2025-10-11 19:18:52     2025-10-11 19:19:08     GROUP_BY,FILT
ER

Success!

Job Stats (time in seconds):
JobId   Maps    Reduces MaxMapTime      MinMapTime      AvgMapTime      MedianMapTime   MaxRe
duceTime        MinReduceTime   AvgReduceTime   MedianReducetime        Alias   Feature Outpu
ts
job_1756381367829_0536  1       1       1       1       1       1       1       1       1   1
all_grouped,corrupt_records,corrupted_records,group_corrupt     GROUP_BY,COMBINER       hdfs:
//msba-hadoop-name:9000/tmp/temp-910750932/tmp-2102845625,

Input(s):
Successfully read 5 records (446 bytes) from: "/home/13student13/input/sample_corrupt.txt"

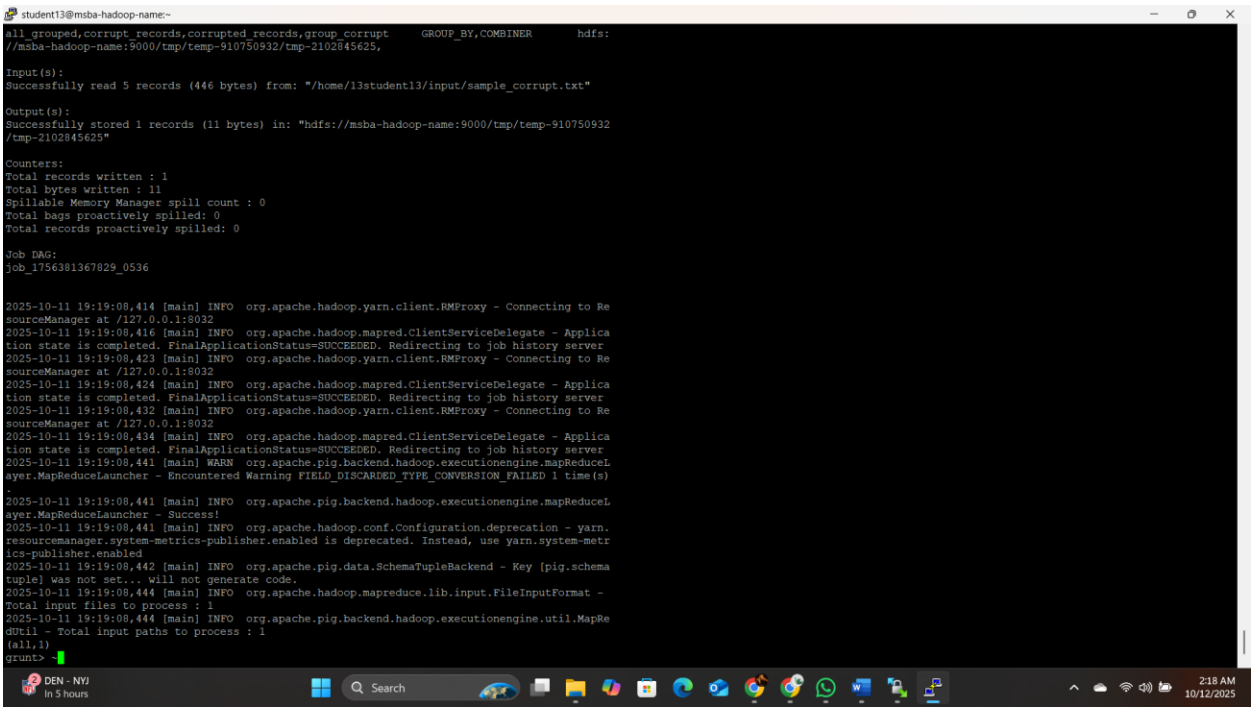
Output(s):
Successfully stored 1 records (11 bytes) in: "hdfs://msba-hadoop-name:9000/tmp/temp-910750932
/tmp-2102845625"

Counters:
Total records written : 1
Total bytes written : 11
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1756381367829_0536


2025-10-11 19:19:08,414 [main] INFO  org.apache.hadoop.yarn.client.RMProxy - Connecting to Re
sourceManager at /127.0.0.1:8032
2025-10-11 19:19:08,416 [main] INFO  org.apache.hadoop.mapred.ClientServiceDelegate - Applica
tion state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
```

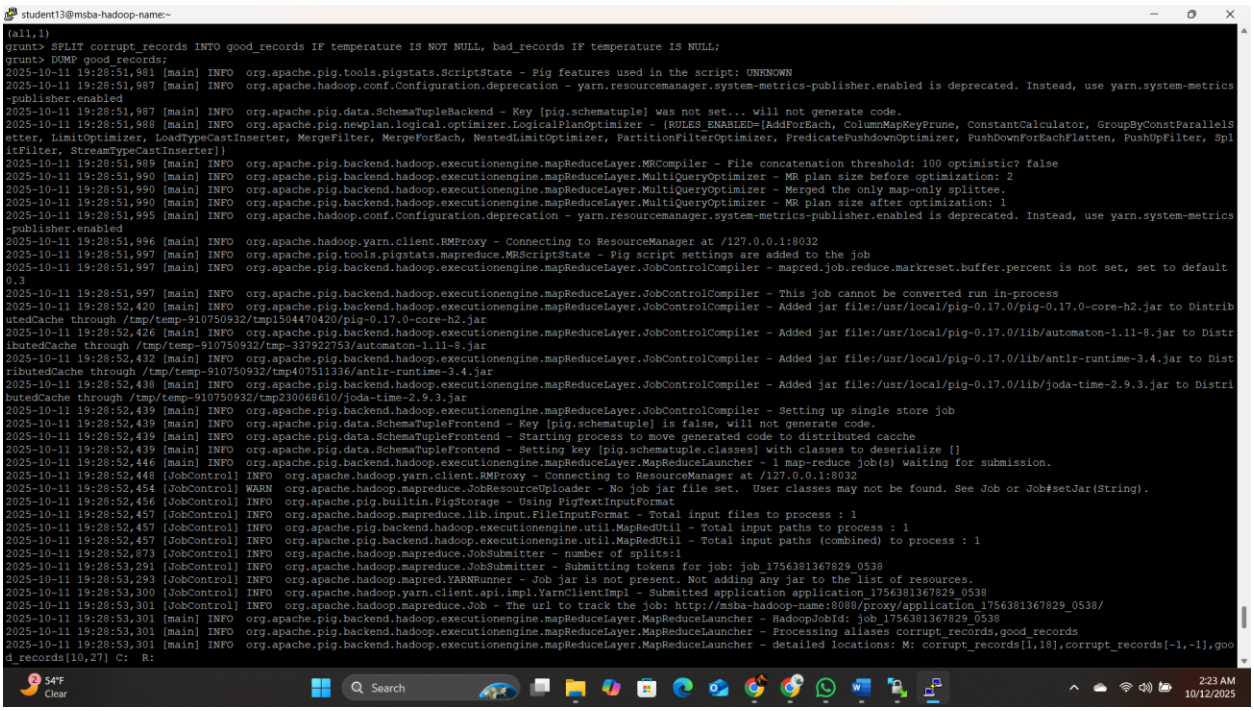
DEN - NYJ
In 5 hours
Search
2:18 AM
10/12/2025

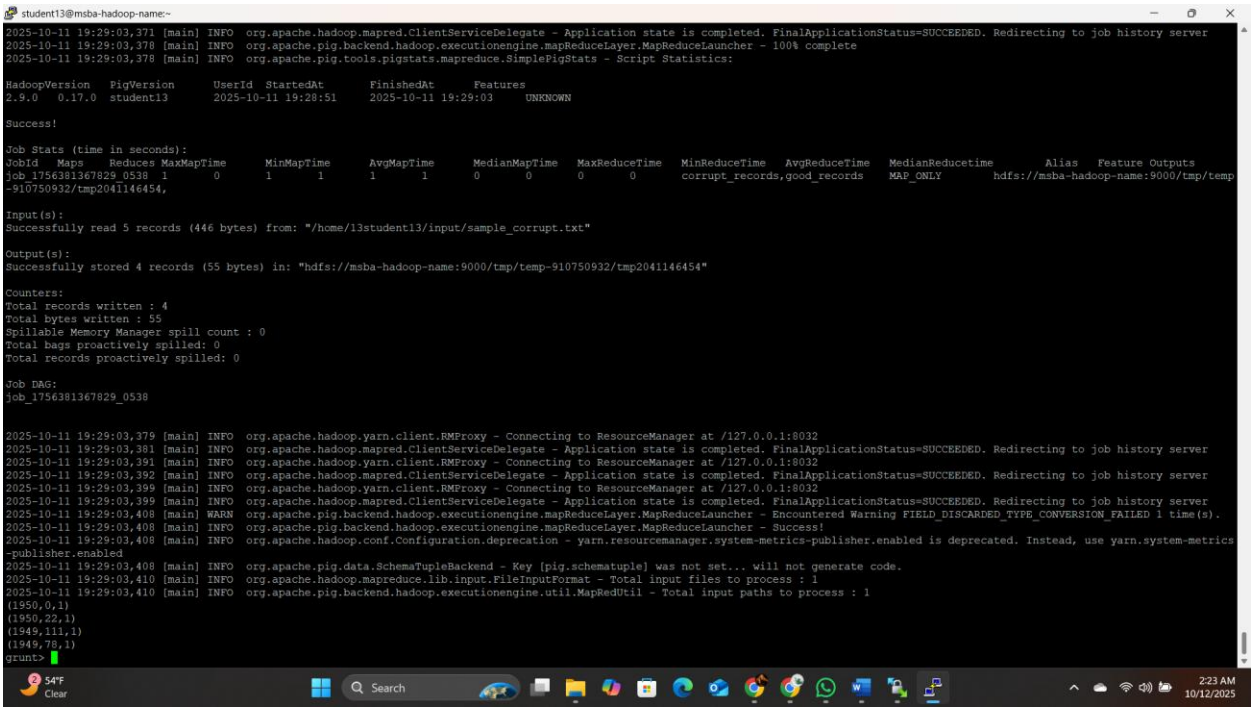


e. Splitting and displaying good and bad records separately -

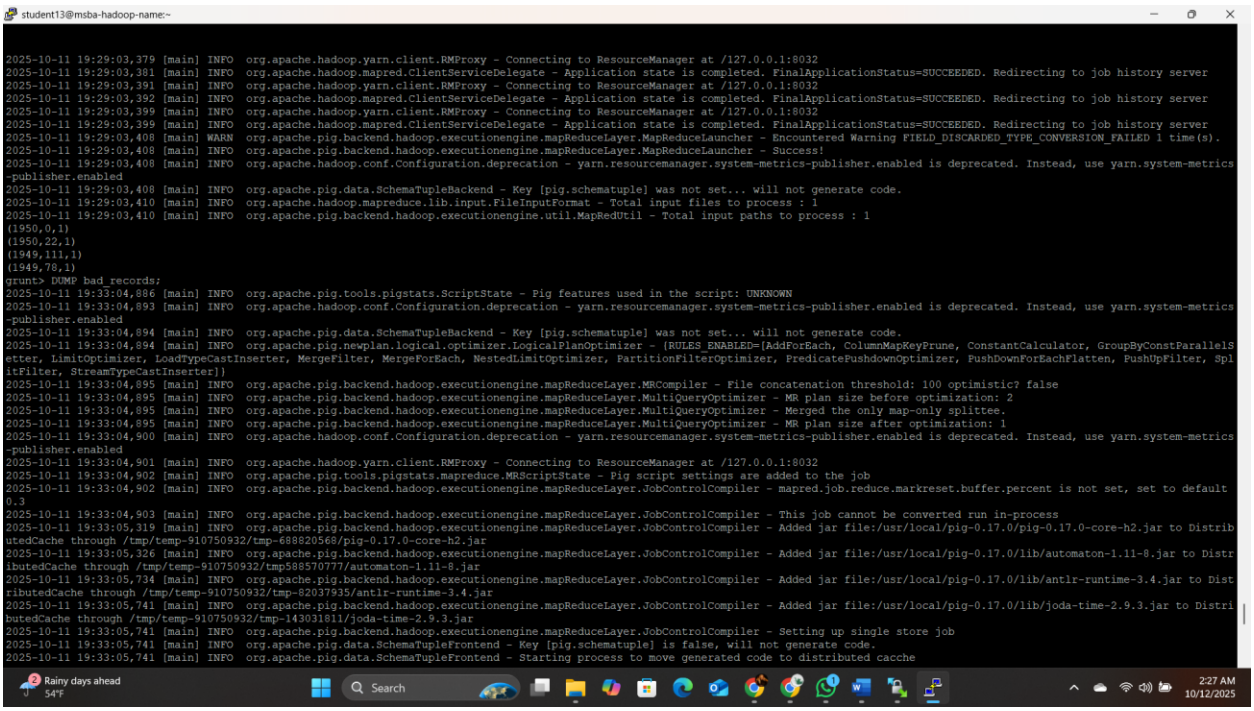
grunt> SPLIT corrupt_records INTO good_records IF temperature IS NOT NULL, bad_records IF temperature IS NULL;

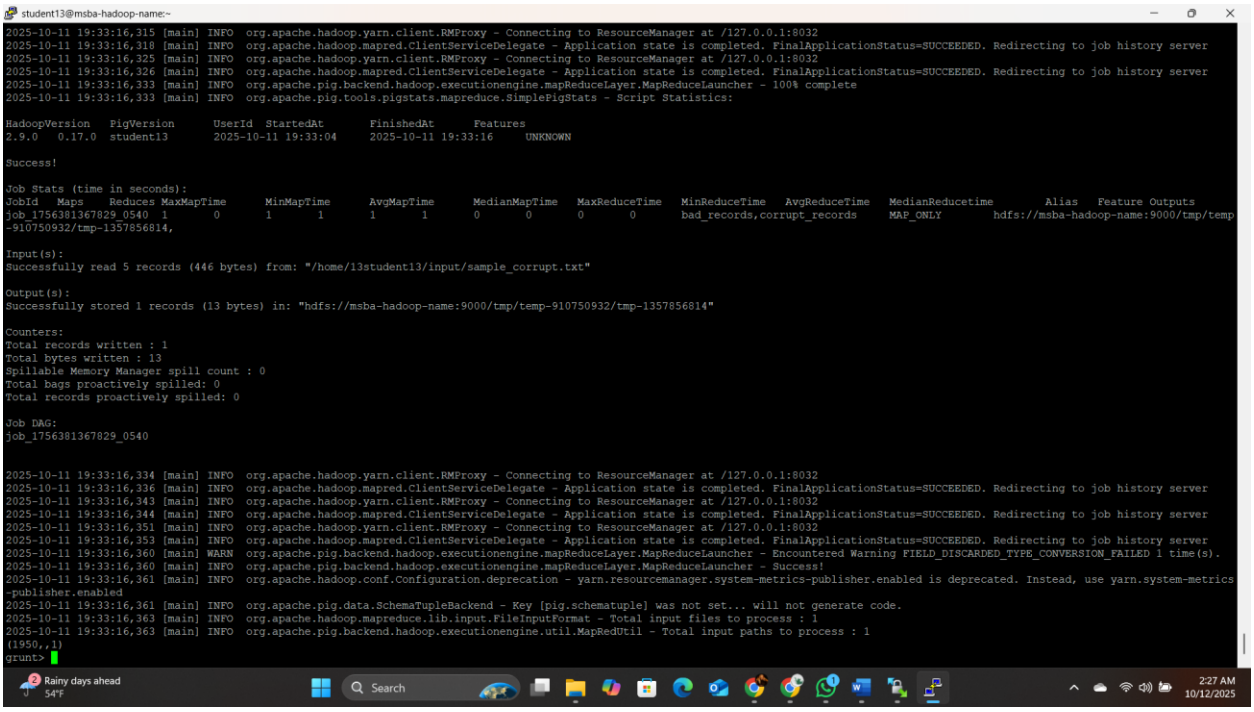
grunt> DUMP good_records;





DUMP bad_records;


student13@msba-hadoop-name:~
2025-10-11 19:29:03,379 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:29:03,381 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:29:03,391 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:29:03,392 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:29:03,399 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:29:03,399 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:29:03,408 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 1 time(s).
2025-10-11 19:29:03,408 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2025-10-11 19:29:03,408 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:29:03,408 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-10-11 19:29:03,410 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2025-10-11 19:29:03,410 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(1950,0,1)
(1950,22,1)
(1949,111,1)
(1949,78,1)
grunt> DUMP bad_records;
2025-10-11 19:33:04,886 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNKNOWN
2025-10-11 19:33:04,893 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:33:04,894 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-10-11 19:33:04,894 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2025-10-11 19:33:04,895 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2025-10-11 19:33:04,895 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 2
2025-10-11 19:33:04,895 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - Merged the only map-only splittee.
2025-10-11 19:33:04,895 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2025-10-11 19:33:04,900 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:33:04,901 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:33:04,902 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2025-10-11 19:33:04,902 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2025-10-11 19:33:04,903 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2025-10-11 19:33:05,319 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/pig-0.17.0-core-h2.jar to DistributedCache through /tmp/temp-910750932/tmp-688820568/pig-0.17.0-core-h2.jar
2025-10-11 19:33:05,326 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp-910750932/tmp588570777/automaton-1.11-8.jar
2025-10-11 19:33:05,734 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp-910750932/tmp-82037935/antlr-runtime-3.4.jar
2025-10-11 19:33:05,741 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/joda-time-2.9.3.jar to DistributedCache through /tmp/temp-910750932/tmp-143031811/joda-time-2.9.3.jar
2025-10-11 19:33:05,741 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2025-10-11 19:33:05,741 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2025-10-11 19:33:05,741 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cacche
Rainy days ahead
54°F
Search
2:27 AM
10/12/2025

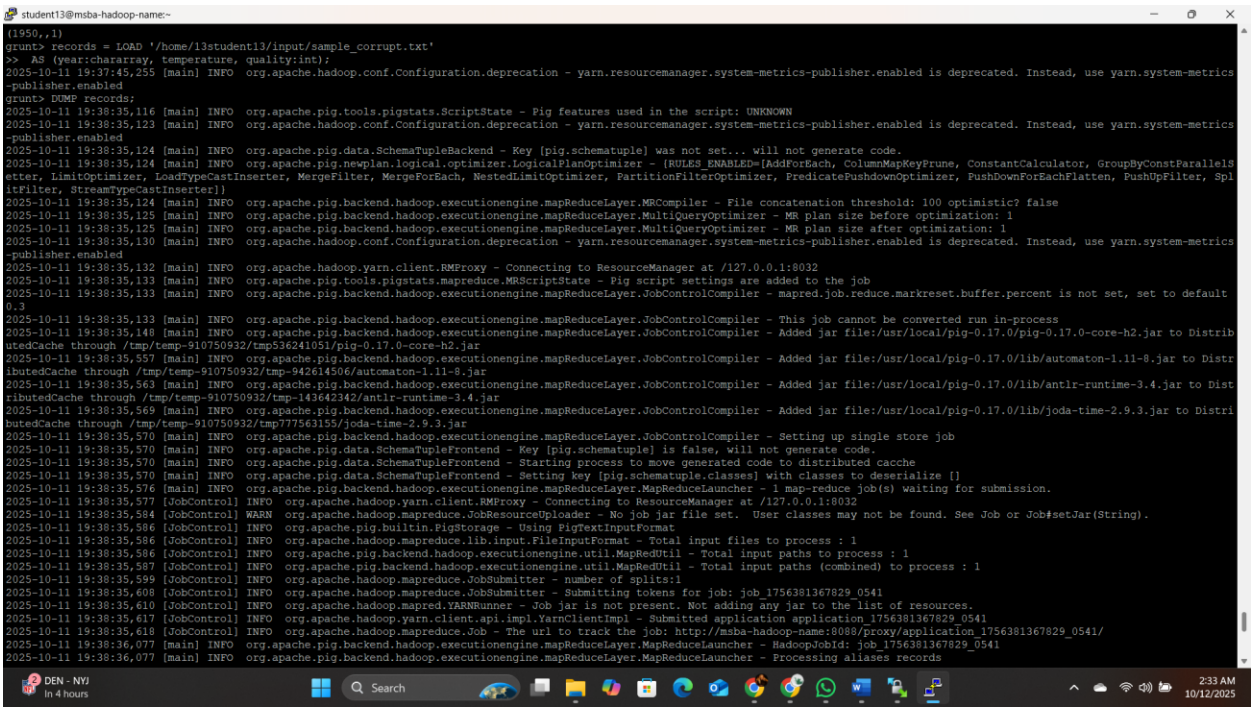

student13@msba-hadoop-name:~
2025-10-11 19:33:16,315 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:33:16,318 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:33:16,325 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:33:16,326 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:33:16,333 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2025-10-11 19:33:16,333 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:
HadoopVersion PigVersion UserId StartedAt FinishedAt Features
2.9.0 0.17.0 student13 2025-10-11 19:33:04 2025-10-11 19:33:16 UNKNOWN
Success!
Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReducetime Alias Feature Outputs
job_1756381367829_0540 1 0 1 1 1 1 0 0 0 0 bad_records,corrupt_records MAP_ONLY hdfs://msba-hadoop-name:9000/tmp/temp-910750932/tmp-1357856814,
Input(s):
Successfully read 5 records (446 bytes) from: "/home/13student13/input/sample_corrupt.txt"
Output(s):
Successfully stored 1 records (13 bytes) in: "hdfs://msba-hadoop-name:9000/tmp/temp-910750932/tmp-1357856814"
Counters:
Total records written : 1
Total bytes written : 13
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0
Job DAG:
job_1756381367829_0540
2025-10-11 19:33:16,334 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:33:16,336 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:33:16,343 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:33:16,344 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:33:16,351 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:33:16,353 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:33:16,360 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Encountered Warning FIELD_DISCARDED_TYPE_CONVERSION_FAILED 1 time(s).
2025-10-11 19:33:16,360 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2025-10-11 19:33:16,361 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:33:16,361 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-10-11 19:33:16,363 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2025-10-11 19:33:16,363 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(1950,,1)
grunt>
Rainy days ahead
54°F
Search
2:27 AM
10/12/2025

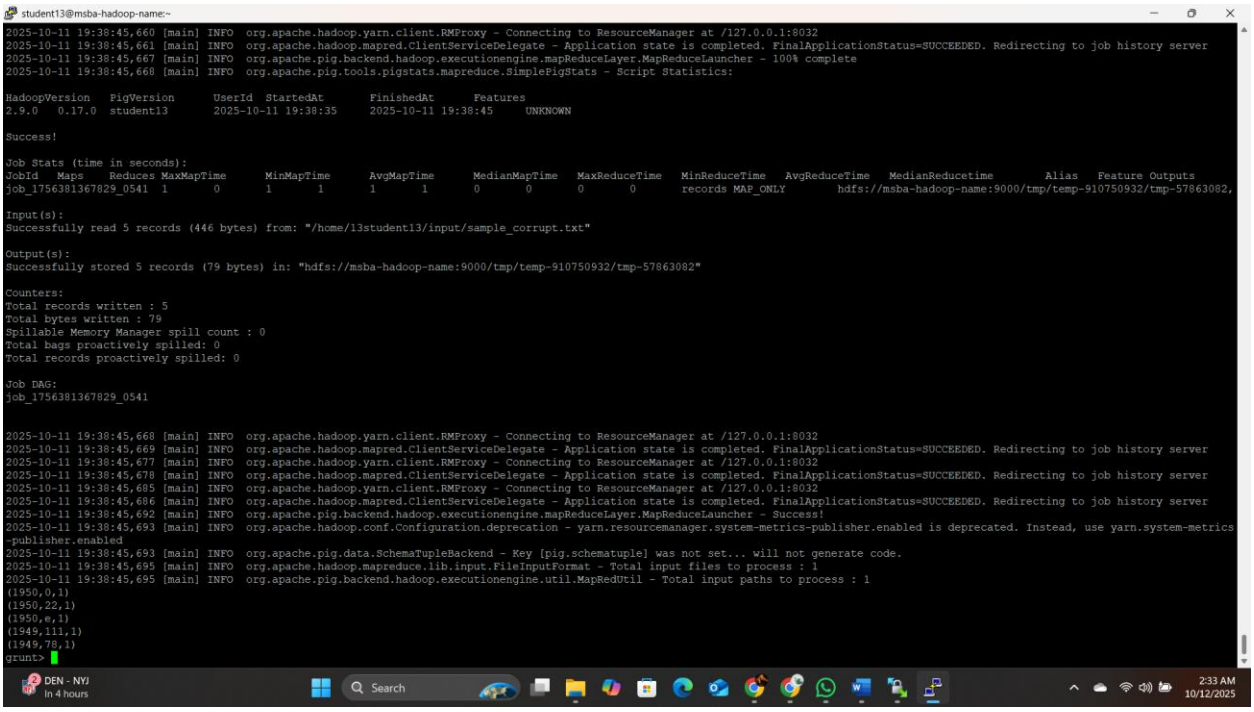
f. Filter records by temperature and quality -

Not defining the data type of temperature:

```
grunt> records = LOAD '/home/13student13/input/sample_corrupt.txt'
>> AS (year:chararray, temperature, quality:int);
```

```
grunt> DUMP records;
```


student13@msba-hadoop-name:~
(1950,,1)
grunt> records = LOAD '/home/13student13/input/sample_corrupt.txt'
>> AS (year:chararray, temperature, quality:int);
2025-10-11 19:37:45,255 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
grunt> DUMP records;
2025-10-11 19:38:35,116 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: UNKNOWN
2025-10-11 19:38:35,123 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:38:35,124 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-10-11 19:38:35,124 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, ConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFilter, StreamTypeCastInserter]}
2025-10-11 19:38:35,124 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2025-10-11 19:38:35,125 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 1
2025-10-11 19:38:35,125 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 1
2025-10-11 19:38:35,130 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:38:35,132 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:38:35,133 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2025-10-11 19:38:35,133 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2025-10-11 19:38:35,133 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - This job cannot be converted run in-process
2025-10-11 19:38:35,148 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/pig-0.17.0-core-h2.jar to DistributedCache through /tmp/temp-910750932/tmp536241051/pig-0.17.0-core-h2.jar
2025-10-11 19:38:35,557 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/automaton-1.11-8.jar to DistributedCache through /tmp/temp-910750932/tmp-94261450 6/automaton-1.11-8.jar
2025-10-11 19:38:35,563 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/antlr-runtime-3.4.jar to DistributedCache through /tmp/temp-910750932/tmp-143642342/antlr-runtime-3.4.jar
2025-10-11 19:38:35,569 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Added jar file:/usr/local/pig-0.17.0/lib/joda-time-2.9.3.jar to DistributedCache through /tmp/temp-910750932/tmp777563155/joda-time-2.9.3.jar
2025-10-11 19:38:35,570 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting up single store job
2025-10-11 19:38:35,570 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Key [pig.schematuple] is false, will not generate code.
2025-10-11 19:38:35,570 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Starting process to move generated code to distributed cacche
2025-10-11 19:38:35,570 [main] INFO org.apache.pig.data.SchemaTupleFrontend - Setting key [pig.schematuple.classes] with classes to deserialize []
2025-10-11 19:38:35,576 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1 map-reduce job(s) waiting for submission.
2025-10-11 19:38:35,577 [JobControl] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:38:35,584 [JobControl] WARN org.apache.hadoop.mapreduce.JobResourceUploader - No job jar file set. User classes may not be found. See Job or Job#setJar(String).
2025-10-11 19:38:35,586 [JobControl] INFO org.apache.pig.builtin.PigStorage - Using PigTextInputFormat
2025-10-11 19:38:35,586 [JobControl] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2025-10-11 19:38:35,586 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
2025-10-11 19:38:35,587 [JobControl] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths (combined) to process : 1
2025-10-11 19:38:35,599 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - number of splits:1
2025-10-11 19:38:35,608 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1756381367829_0541
2025-10-11 19:38:35,610 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2025-10-11 19:38:35,617 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1756381367829_0541
2025-10-11 19:38:35,618 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://msba-hadoop-name:8088/proxy/application_1756381367829_0541/
2025-10-11 19:38:36,077 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - HadoopJobId: job_1756381367829_0541
2025-10-11 19:38:36,077 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases records
DEN - NYJ
In 4 hours
Search
2:33 AM
10/12/2025

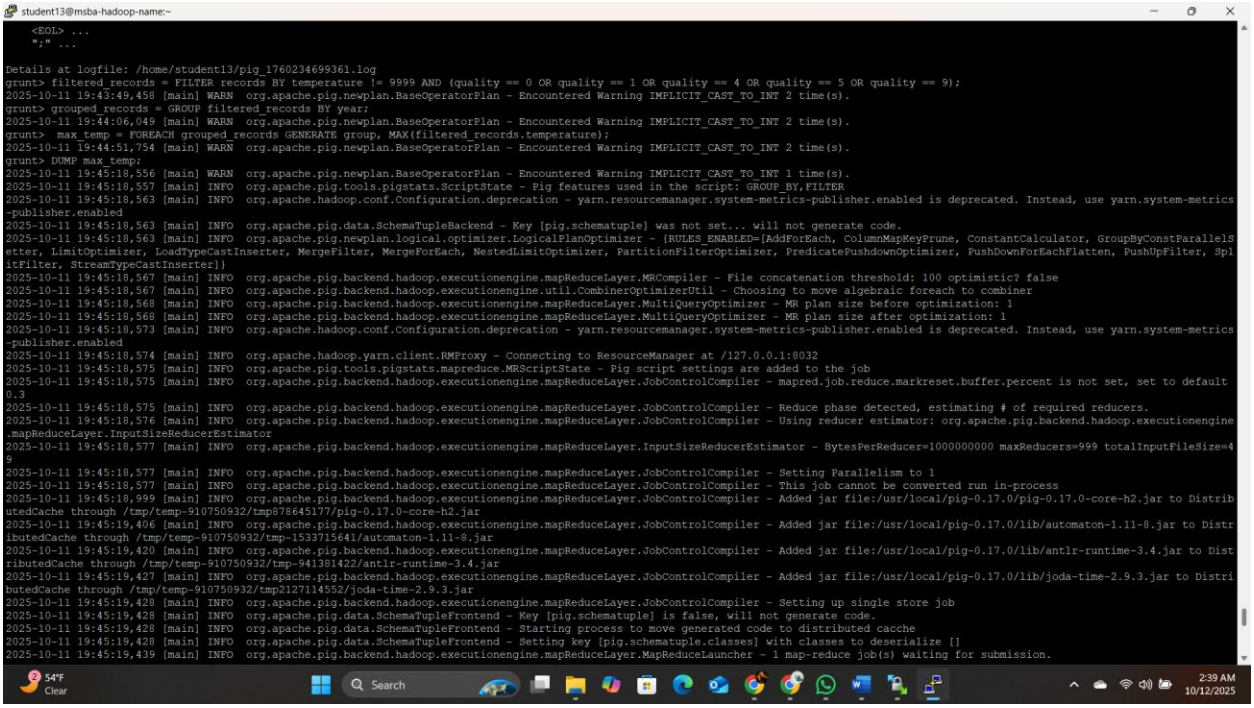

student13@msba-hadoop-name:~
2025-10-11 19:38:45,660 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:38:45,661 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:38:45,667 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2025-10-11 19:38:45,668 [main] INFO org.apache.pig.tools.pigstats.mapreduce.SimplePigStats - Script Statistics:
HadoopVersion PigVersion UserId StartedAt FinishedAt Features
2.9.0 0.17.0 student13 2025-10-11 19:38:35 2025-10-11 19:38:45 UNKNOWN
Success!
Job Stats (time in seconds):
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime AvgReduceTime MedianReducetime Alias Feature Outputs
job_1756381367829_0541 1 0 1 1 1 1 0 0 0 0 records MAP_ONLY hdfs://msba-hadoop-name:9000/tmp/temp-910750932/tmp-57863082,
Input(s):
Successfully read 5 records (446 bytes) from: "/home/13student13/input/sample_corrupt.txt"
Output(s):
Successfully stored 5 records (79 bytes) in: "hdfs://msba-hadoop-name:9000/tmp/temp-910750932/tmp-57863082"
Counters:
Total records written : 5
Total bytes written : 79
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0
Job DAG:
job_1756381367829_0541
2025-10-11 19:38:45,668 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:38:45,669 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:38:45,677 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:38:45,678 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:38:45,685 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /127.0.0.1:8032
2025-10-11 19:38:45,686 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2025-10-11 19:38:45,692 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2025-10-11 19:38:45,693 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - yarn.resourcemanager.system-metrics-publisher.enabled is deprecated. Instead, use yarn.system-metrics-publisher.enabled
2025-10-11 19:38:45,693 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will not generate code.
2025-10-11 19:38:45,695 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input files to process : 1
2025-10-11 19:38:45,695 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths to process : 1
(1950,0,1)
(1950,22,1)
(1950,e,1)
(1949,111,1)
(1949,78,1)
grunt>
DEN - NYJ
In 4 hours
Search
2:33 AM
10/12/2025

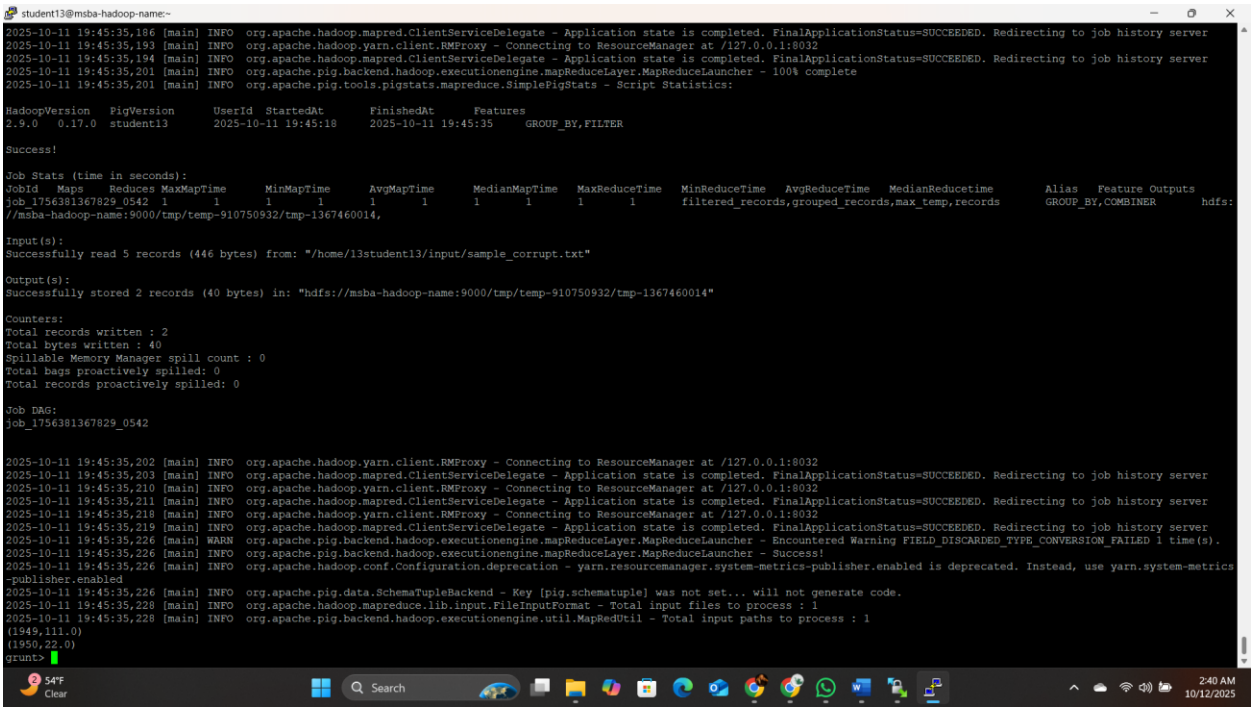
```
grunt> filtered_records = FILTER records BY temperature != 9999 AND (quality == 0
OR quality == 1 OR quality == 4 OR quality == 5 OR quality == 9);

grunt> grouped_records = GROUP filtered_records BY year;

grunt> max_temp = FOREACH grouped_records GENERATE group,
MAX(filtered_records.temperature);


grunt> DUMP max_temp;
```





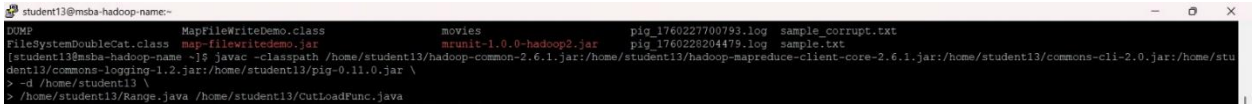
Q3: Work “A Load UDF” example on pages 396 and 397. Type out all the commands in each step of the process and print out a screenshot(s) of all the steps and the final results in CSUEB Hadoop.

Note. 1) This example also has Range.java, which is located in the same folder as

CutLoadFunc.java. 2) Need pig-0.11.0.jar and commons-logging-1.2.jar for compilation. The two jars are available in Useful Jars folder on Canvas.

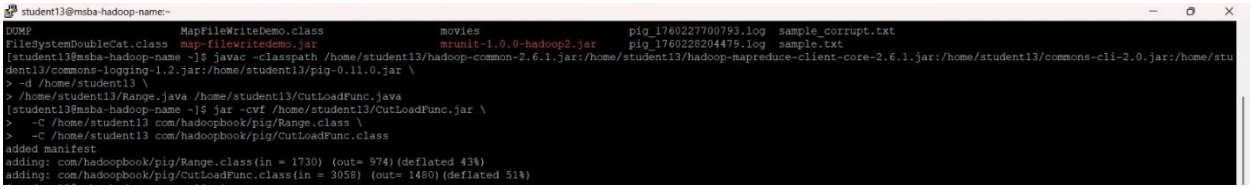
Step 1: Compile the Java File CutLoadFunc and Range into class files

```
javac -classpath /home/student13/hadoop-common-2.6.1.jar:/home/student13/hadoop-
mapreduce-client-core-2.6.1.jar:/home/student13/commons-cli-
2.0.jar:/home/student13/commons-logging-1.2.jar:/home/student13/pig-0.11.0.jar \
-d /home/student13 \
/home/student13/Range.java /home/student13/CutLoadFunc.java
```



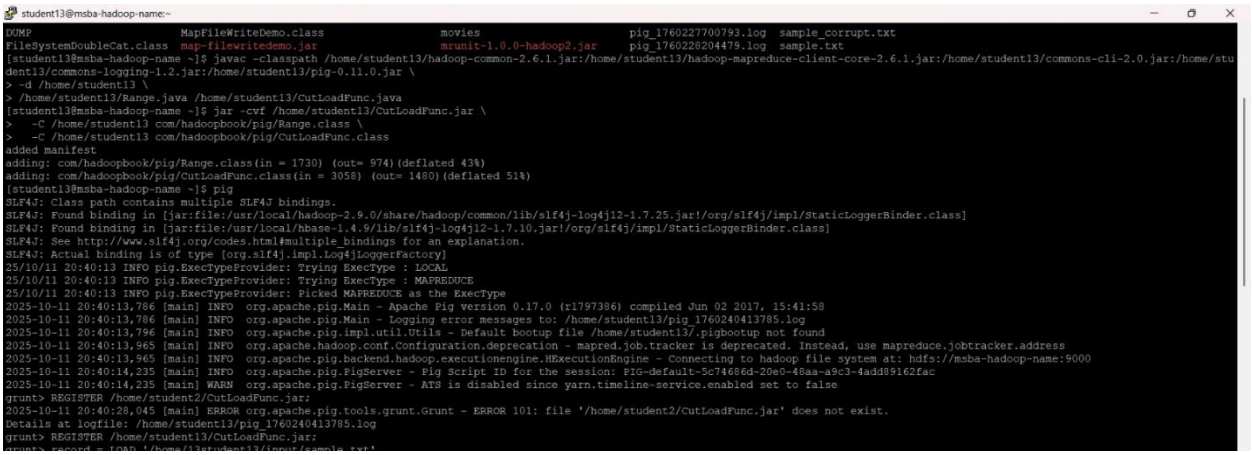
Step 2: Compile the Class File CutLoadFunc and Range as CutLoadFunc.jar

```
jar -cvf /home/student13/CutLoadFunc.jar \
  -C /home/student13 com/hadoopbook/pig/Range.class \
  -C /home/student13 com/hadoopbook/pig/CutLoadFunc.class
```



Step 3: Register the CutLoadFunc.jar

```
REGISTER /home/student13/CutLoadFunc.jar;
```

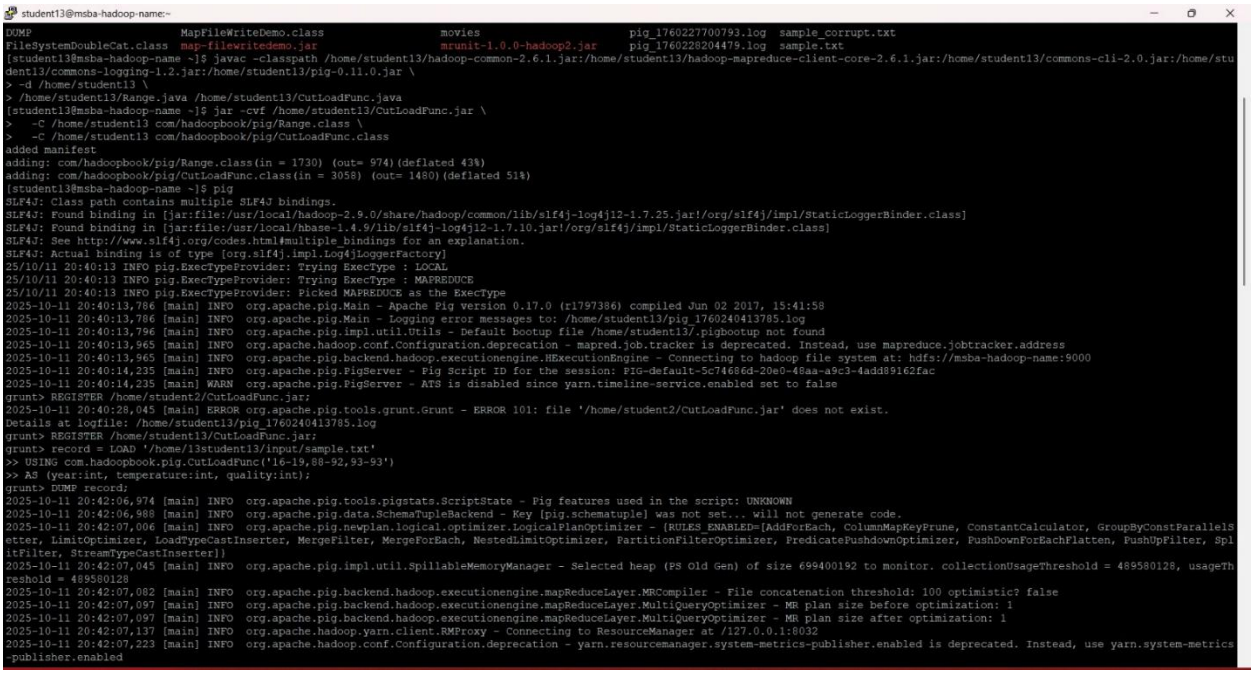


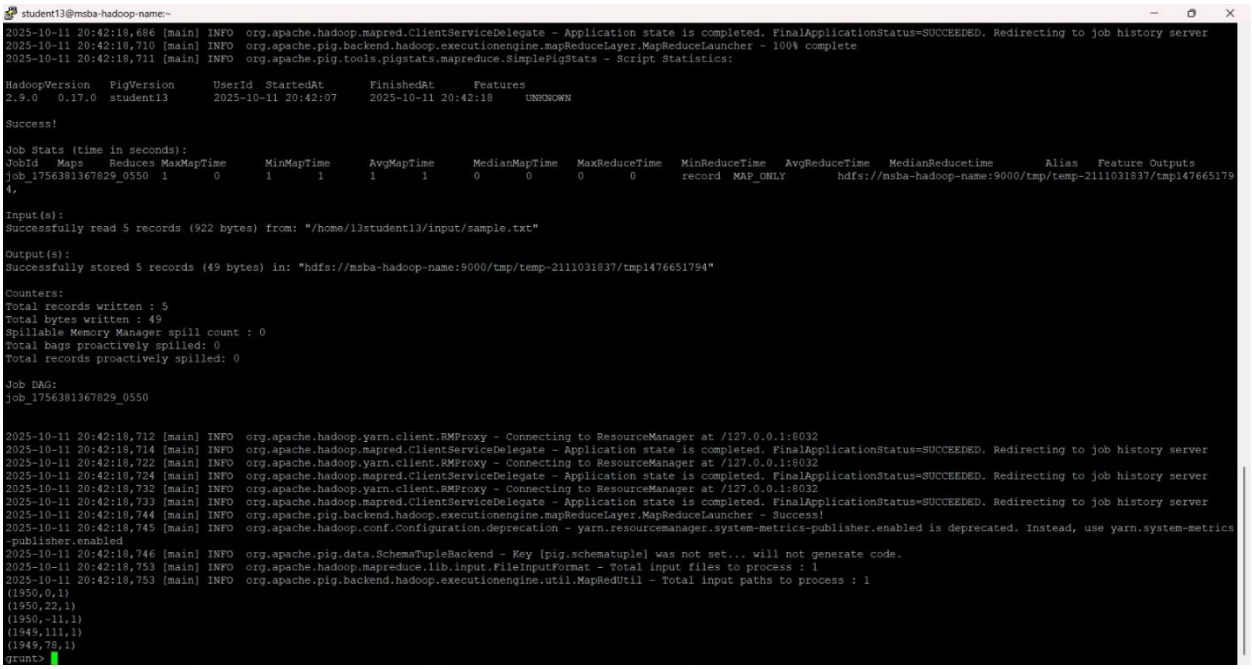
Step 4: Run the CutLoadFunc UDF

grunt> record = LOAD '/home/13student13/input/sample.txt'

>> USING com.hadoopbook.pig.CutLoadFunc('16-19,88-92,93-93')

>> AS (year:int, temperature:int, quality:int);





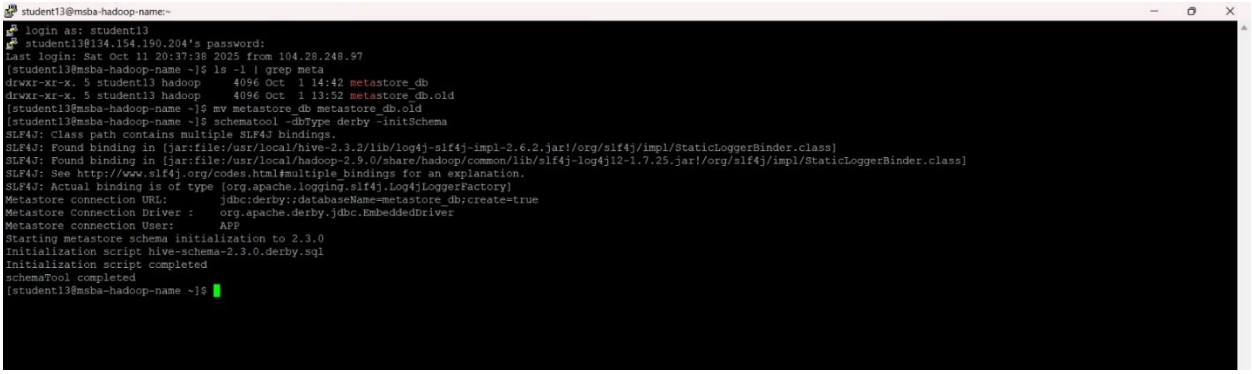
Q4: Work the four examples on pages 416 and 417, that is, from the title “An Example” to the title “Running Hive.” Type out all the commands in each step of the process and print out a screenshot(s) of all the steps and the final results in CSUEB Hadoop.

Step 1: Initializing db in hive Check if files or directories related to “meta” (like metastore_db) exist in the current directory

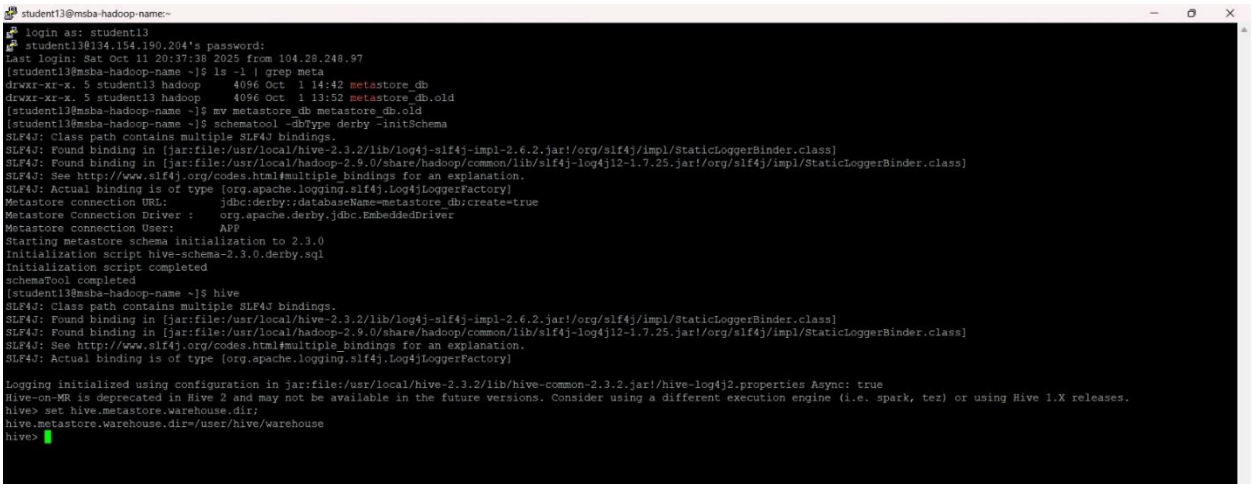
ls -l | grep meta



Back up or move the existing Hive metastore database directory so that a new one can be created safely.
mv metastore_db metastore_db.old
Recreate the Hive metastore schema in a fresh Derby database.
schematool -dbType derby -initSchema

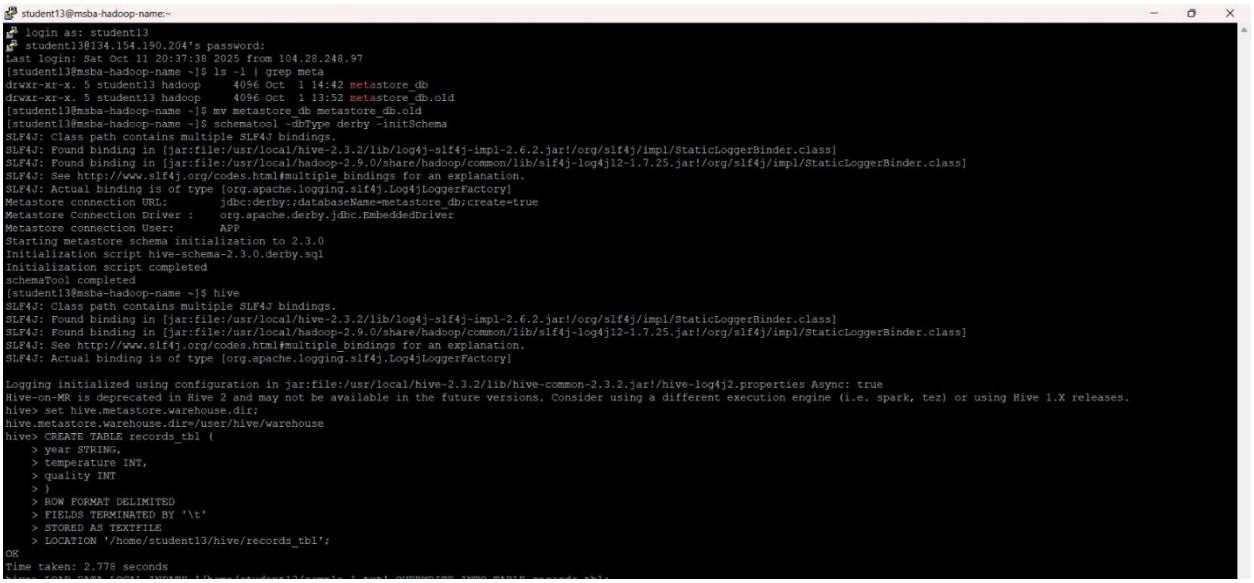


Step 2: Set the value of the Hive configuration property
set hive.metastore.warehouse.dir;



Step 3: Creating ‘records_tbl’ table:

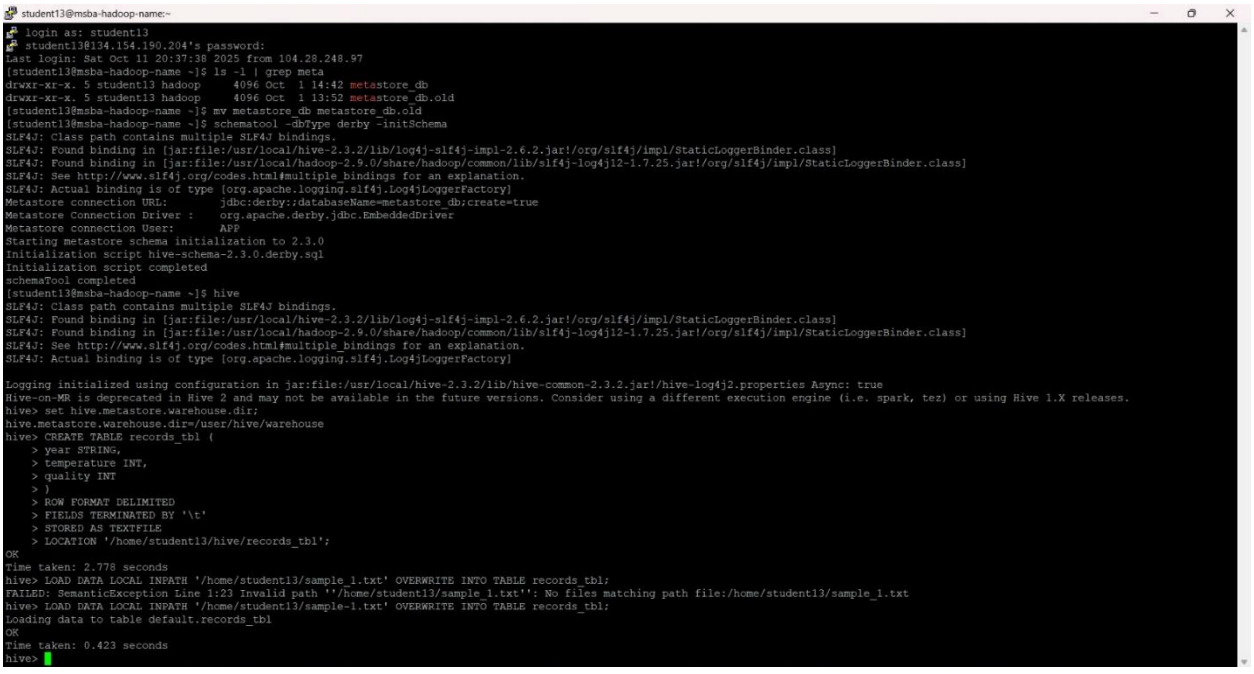
```
CREATE TABLE records_tbl (
year STRING, temperature INT,
quality INT ) ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t'
STORED AS TEXTFILE
LOCATION '/home/student13/hive/records_tbl';
```



Step 4: Loading the Data to the ‘records_tbl’ table with sample_1.txt

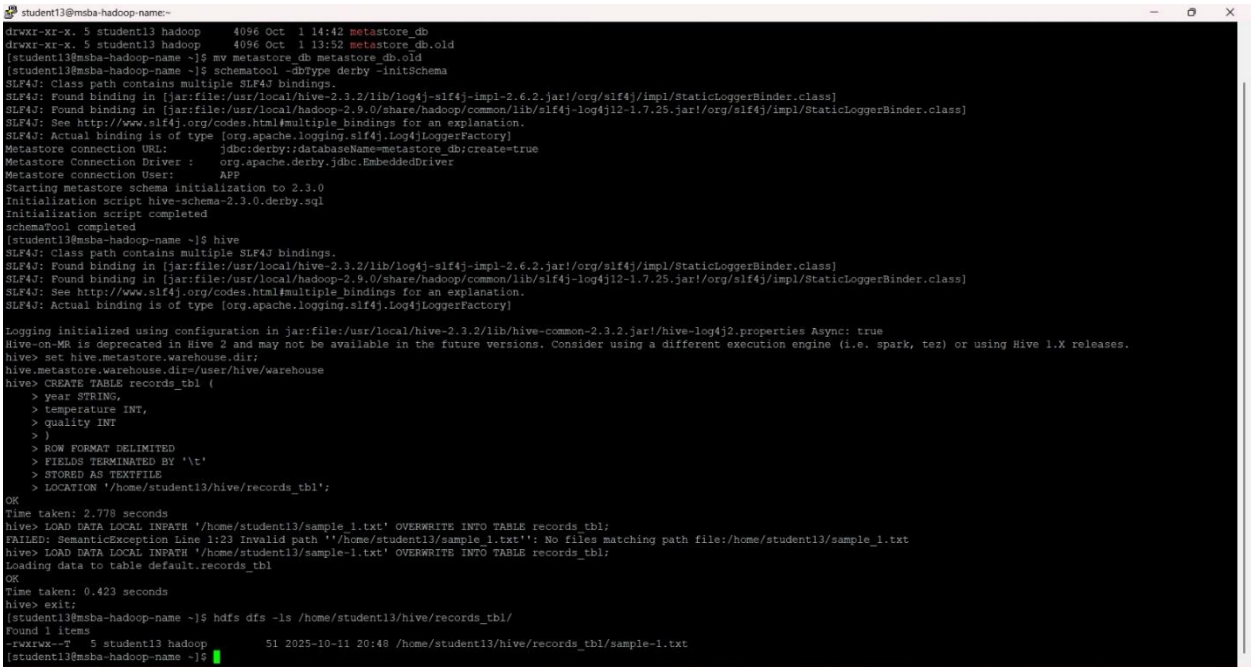
(sample_1.txt is same as sample.txt from ncdc > micro-tab)

LOAD DATA LOCAL INPATH '/home/student13/sample_1.txt' OVERWRITE INTO TABLE records_tbl;



Step 5: Checking where the input data file in the ‘records_tbl’ table is located:

hdfs dfs -ls /home/student13/hive/records_tbl/



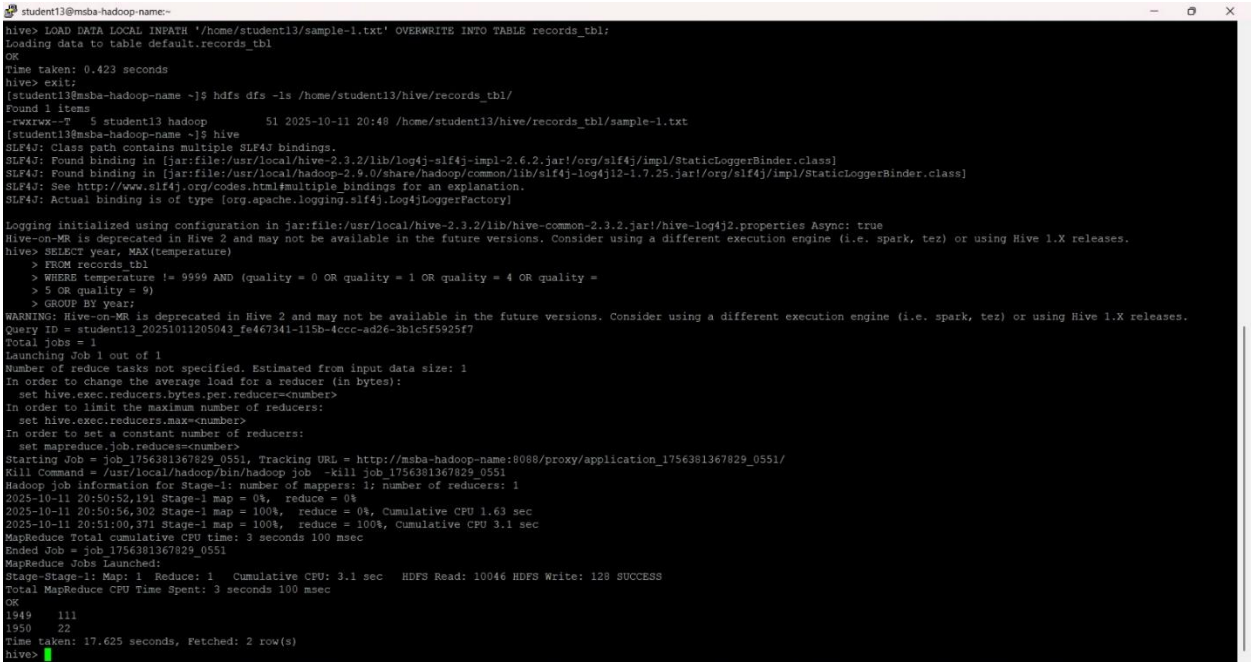
Step 6: Running the query on ‘records_tbl’ table

SELECT year, MAX(temperature)

FROM records_tbl

WHERE temperature != 9999 AND (quality = 0 OR quality = 1 OR quality = 4 OR quality = 5 OR quality = 9)

GROUP BY year;


student13@msba-hadoop-name:~
hive> LOAD DATA LOCAL INPATH '/home/student13/sample-1.txt' OVERWRITE INTO TABLE records_tbl;
Loading data to table default.records_tbl
OK
Time taken: 0.423 seconds
hive> exit;
[student13@msba-hadoop-name ~]$ hdfs dfs -ls /home/student13/hive/records_tbl/
Found 1 items
-rwxrwx--T   5 student13 hadoop   51 2025-10-11 20:48 /home/student13/hive/records_tbl/sample-1.txt
[student13@msba-hadoop-name ~]$ hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/usr/local/hive-2.3.2/lib/log4j-slf4j-impl-2.6.2.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/local/hadoop-2.9.0/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
Logging initialized using configuration in jar:file:/usr/local/hive-2.3.2/lib/hive-common-2.3.2.jar!/hive-log4j2.properties Async: true
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
hive> SELECT year, MAX(temperature)
    > FROM records_tbl
    > WHERE temperature != 9999 AND (quality = 0 OR quality = 1 OR quality = 4 OR quality =
    > 5 OR quality = 9)
    > GROUP BY year;
WARNING: Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.
Query ID = student13_20251011205043_fe467341-115b-4ccc-ad26-3b1c5f5925f7
Total jobs = 1
Launching Job 1 out of 1
Number of reduce tasks not specified. Estimated from input data size: 1
In order to change the average load for a reducer (in bytes):
  set hive.exec.reducers.bytes.per.reducer=<number>
In order to limit the maximum number of reducers:
  set hive.exec.reducers.max=<number>
In order to set a constant number of reducers:
  set mapreduce.job.reduces=<number>
Starting Job = job_1756381367829_0551, Tracking URL = http://msba-hadoop-name:8088/proxy/application_1756381367829_0551/
Kill Command = /usr/local/hadoop/bin/hadoop job -kill job_1756381367829_0551
Hadoop job information for Stage-1: number of mappers: 1; number of reducers: 1
2025-10-11 20:50:52,191 Stage-1 map = 0%, reduce = 0%
2025-10-11 20:50:56,302 Stage-1 map = 100%, reduce = 0%, Cumulative CPU 1.63 sec
2025-10-11 20:51:00,371 Stage-1 map = 100%, reduce = 100%, Cumulative CPU 3.1 sec
MapReduce Total cumulative CPU time: 3 seconds 100 msec
Ended Job = job_1756381367829_0551
MapReduce Jobs Launched:
Stage-Stage-1: Map: 1 Reduce: 1   Cumulative CPU: 3.1 sec   HDFS Read: 10046 HDFS Write: 128 SUCCESS
Total MapReduce CPU Time Spent: 3 seconds 100 msec
OK
1949   111
1950   22
Time taken: 17.625 seconds, Fetched: 2 row(s)
hive>